MILITARY SPECIFICATION
DUAL MODE TRACKER
FOR
BOMBING SET, ANGLE RATE AN/ASB-19(V)

This specification is approved for use by the Naval Air System Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, functional, and test requirements for two configurations of an airborne dual mode tracker for the AN/ASB-19(V) Angle Rate Bombing Set (ARBS). The Dual Mode Tracker (DMT) will herein be referred to as "the tracker." The two configurations are differentiated by the following nomenclature:

a. for A-4M aircraft AN/ASB-19(V)1
b. for AV-8B aircraft AN/ASB-19(V)2

Note: The Signal Data Converter and Heat Exchanger Hose Assemblies differ for A-4M and AV-8B; see 1.2 (a) and (b) below.

1.2 Classification. The tracker equipment covered by this specification shall consist of the following items:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ, 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1280

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Type Designation	Part Number	Applicable Paragraph
a. A-4M:		
Receiver-Processor R-2028/ASB-19(V)	3366021-100	3.4.1.1
Converter, Signal Data CV-3393/ASB-19(V)	3366022-100	3.4.1.2
Heat Exchanger HD-1005/ASB-19(V)	3366030-100	3.4.1.3
Heat Exchanger Hose Assy. (two). Related Assy.	3491400	
b. AV-8B:		
Receiver-Processor R-2028/ASB-19(V)	3366021-100	3.4.1.1
Converter, Signal Data CV-3693/ASB-19(V)	3366022-101	3.4.1.2
Heat Exchanger HD-1005/ASB-19(V)	3366030-100	3.4.1.3
Heat Exchanger Hose Assy. (Two Pieces One of Each) Related Assy.	3366801-1 and 3366801-2	

Paragraphs that apply to both A-4M and AV-8B are marked either A-4M and AV-8B, or, remain unmarked. If applicable to A-4M only, then the paragraph will be marked A-4M Only; If applicable to AV-8B only, then the paragraph will be marked AV-8B Only.

1.3 Associated Equipment. The tracker operates with the following equipment in accordance with MIL-A-85055:

a. A-4M Associated Equipment:

Item	Specification/Part Number
Weapon Delivery Computer (WDC)	MIL-W-85057
Control Unit (CU) Subsystem	MIL-C-85058
TV Monitor	IP-936-AXQ
Attitude Reference System	AN/AJB-3A

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b. AV-8B Associated Equipment:

Mission Computer (MC)	AS-4197(V) AMM1/AN/AYK-14(V)
Inertial Navigation Set, AN/ASN-130	PS74-870082/AN/ASN-130
Display Processor	PS75-870105
Stability Augmentation and Attitude Hold System (SAAHS)	PS75-874051

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. Unless otherwise specified herein, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-M-7793	Meter, Time Totalizing
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment; Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Ground Support Equipment
MIL-A-85055	Specification for A-4M/AV-8B Bombing Set, Angle Rate AN/ASB-19(V)
MIL-W-85057	Weapon Delivery Computer for the Bombing Set, Angle Rate AN/ASB-19(V)
MIL-C-85058	Control Unit Subsystem for the Bombing Set, Angle Rate AN/ASB-19(V)
MIL-T-85115	Tracker and Target Characteristics and Sensitivity Requirements of the Dual Mode Tracker for the Bombing Set, Angle Rate AN/ASB-19(V) (U)

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Naval Air Systems Command (Code Ident 30003)

AS-4197	Computer Set, Standard Airborne AN/AYK-14(V)
SD-563-3-1	Detail Specification for the Design of Model AV-8B Aircraft

STANDARDS

Military

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Test Exponential Distributions
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment

2.1.1 Availability of Documents. Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DRAWINGS

Hughes Aircraft Company (Code Ident 82577)

A-4M

3366000-310	Wire List
3366000-350	Electrical Interface Data
3366020-100	Dual Mode Tracker
3366021-500	Installation Control Drawing, Receiver-Processor

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3366022-500	Installation Control Drawing, Signal Data Converter
3366030-500	Installation Control Drawing, Heat Exchanger Unit
AV-8B	
3366020-101	Dual Mode Tracker
3366021-500	Installation Control Drawing, Receiver-Processor
3366022-500	Installation Control Drawing, Signal Data Converter
3366030-500	Installation Control Drawing, Heat Exchanger Unit
3366000-350	Electrical Interface Data (A-4M drawing applies to AV-8B only in so far as required by Paragraph 3.4.5.3.7 Instrumentation Interface)

SPECIFICATIONS

McDonnell Douglas Corporation (Code Ident. 18355)

MDC A5232	AV-8B Avionics Multiplex Bus Design Specification
75-870105	Display Processor (Harrier)
74-870082	Navigation Set, Inertial AN/ASN-130
75J875090	Wiring Diagram Basic Schematic AV-8B
75J875900	Interface Control Document

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

3.2 First Article. When specified (see 6.2), the contractor shall furnish sample units(s) for first article inspection and approval.

3.2.1 Parts and Materials. Selection of parts and materials shall be in accordance with the requirements of MIL-A-85055.

3.3 Design and Construction. The tracker shall conform with all the applicable requirements of MIL-A-85055 for design, construction and workmanship, except as otherwise specified herein or the contract.

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3.3.1 Physical Characteristics. The tracker shall have the physical characteristics specified herein.

3.3.1.1 Total Weight. The total weight of the tracker, excluding cables, shall be a minimum consistent with good design and shall not exceed 85 pounds.

3.3.1.2 Dimensions. Dimensions of the tracker components shall be in accordance with Drawings 3366021-500, 3366022-500, and 3366030-500, as applicable.

3.3.2 Reliability. The contractor shall conduct a reliability program in accordance with MIL-A-85055, except as specified herein.

3.3.2.1 Operational Stability. The tracker shall operate with satisfactory performance, continuously or intermittently for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life. The tracker shall have a total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor, and approved by the procuring activity.

3.3.2.3 Reliability in Mean-Time-Between-Failure (MTBF). The tracker, including any built-in-test provisions, shall have a specified (θ_o) MTBF of 500 hours and a minimum acceptable (θ_1) MTBF of 250 hours when tested as specified herein.

3.3.2.4 Time Totalizing Meter. Time totalizing meters shall be included on the tracker to provide an indication of cumulative operating time. The time totalizing meter shall be installed in a location directly visible without requiring removal of any connectors or opening the unit, and be in accordance with MIL-M-7793. The range of the meters shall be 0 to 9,999 hours and they shall not be capable of being reset.

3.3.3 Cabling and Connectors. The cabling and connectors shall be in accordance with MIL-A-85055.

3.3.3.1 Telemetry Capability. A female test connector shall be provided which contains the necessary electrical outputs to allow inter-

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facing with a telemetry system for in-flight or laboratory evaluation of tracker performance. The monitored functions and signals shall be in accordance with the Electrical Interface Data List, Drawing Installation Control Drawing, Signal Data Converter, Drawing 3366022-500. A connector cap shall be provided for use when the connector is not used.

3.3.4 Interchangeability. All Weapon Replaceable Assemblies (WRA) and Shop Replaceable Assemblies (SRA) shall be directly and completely interchangeable with respect to installation, performance, and construction as required in MIL-A-85055. Reiterative harmonization adjustments shall not be required after replacement of SRAs, except for mechanical and electrical alignment of gimbal elements and vidicon focussing adjustments required for the Receiver-Processor.

3.3.5 Electromagnetic Compatibility. The tracker shall conform to MIL-A-85055.

3.3.5.1 Static Electricity. The tracker shall conform to the requirements of MIL-A-85055 for static electricity.

3.3.5.2 Electromagnetic Interference (EMI). The tracker shall conform to the requirements of MIL-A-85055 for EMI.

3.3.6 Maintainability Provisions. The Maintainability Program, built-in-test features, construction and packaging, provisions for test points, and other maintainability parameters shall conform to the maintainability requirements of MIL-A-85055.

3.3.6.1 Fault Isolation Test (FIT). The tracker shall include a FIT capability, to be performed at the organizational level of maintenance. The FIT shall isolate 90 percent of tracker failures to the WRA level for replacement of the faulty WRA within the time-to-repair requirement of MIL-A-85055.

3.3.6.2 Boresight Provisions. The tracker shall include provisions for mechanical alignment/boresighting of Receiver-Processor gimbals and gimbal elements with respect to the reference mounting base.

3.3.7 Nomenclature, Nameplates and Identifications Marking. Nomenclature and serial number assignment, nameplate approval, and identification marking shall be in accordance with MIL-N-18307. Marking shall be in accordance with MIL-STD-130.

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3.3.7.1 Serialization. Each SRA designed for normal intermediate level replacement shall have a permanent serial number. Serial numbers shall be applied at the final assembly of each SRA. In the event a unit is scrapped, the serial number shall not be reused.

3.3.8 Standard Condition. The standard conditions used to establish normal performance characteristics shall be as specified in MIL-A-85055.

3.3.9 Service Conditions. The tracker, while operating or non-operating, shall withstand the natural and induced environments as specified herein, including all logical adverse combinations, with repeated exposure to conditions having cumulative adverse effects.

3.3.9.1 Nonoperating Conditions. The tracker shall be capable of withstanding the temperature, humidity, vibration, and shock as shown in Table I without physical damage or subsequent degradation of performance.

3.3.9.2 Operating Temperature-Altitude. The tracker shall meet the performance requirements of 3.4 and 3.5 when exposed to the temperature-altitude environments shown in Figure 1, as follows:

a. The tracker shall be capable of continuous operation at the extreme limits of Figure 1, Curve A. Continuous operation is defined as indefinite operation within the extremes of Curve A.

b. The tracker shall be capable of performing intermittent operations at the extreme limits of Figure 1, Curve B, and at Point 1. Intermittent operation of the tracker is defined as 45 minutes' operation (any selected mode) within the limits of Curve A, followed by 12 minutes' operation laser spot tracker (LST) (scan mode) at the upper temperature limit of Curve B, followed by 6 minutes' transient operation at Point 1 (track mode) at 70°C (sea level), followed by 12 minutes' operation (any mode) at limits of Curve B (sea level). Thermal shutdown of the tracker is allowed while the intermittent high temperature condition exists. The high temperature recovery period shall not exceed 5 minutes after the temperature condition is returned to 45°C.

c. The tracker shall be capable of performing intermittent operations at the extreme high temperature limits of Figure 1, Curve C, without part failure; but performance degradation may occur. Intermittent operation of the tracker is defined as in (b), except that the limits of

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Table I. DMT Environmental Criteria-Factory to Target

Environment Item	Event – Condition									
	Transportation				Storage		Handling		Aircraft/Mission	
	Truck	Rail	Ship	Air	Airfield	Carrier	Airfield	Carrier	Non-Oper	Operating
Temperature High °C (°F) Low °C (°F)	50 (122) −30 (−22)	50 (122) −30 (−22)	38 (100) −18 (0)	60 (140) −18 (0)	70 (158) −30 (−22)	38 (100) 0 (32)	50 (122) −30 (−22)	30 (86) 0 (32)	85 (185) −54 (−65)	Figure 1 −40 (−40)
Shock °C/min	NEG	NEG	NEG	NEG	NEG	NEG	NEG	NEG	80	
Altitude–feet	NEG	NEG	NEG	50,000	NEG	S/L	NEG	S/L	50,000	S/L to 45,000
Alt rate–ft/min	NEG	NEG	NEG	1,000	N/A	N/A	N/A	N/A	40,000	
Vibration Sinusoidal, g(PK) Freq range, (Hz)	2 g, 5–12 Hz 0.5g, 12–100 Hz	1 g 5–500 Hz	0.7 g 5–70 Hz	0.4 g, 5–40 5 g 40–700	N/A N/A	N/A N/A	N/A N/A	N/A N/A		
Vibration Random–g^2/Hz Freq range, (Hz)	See above				N/A	N/A	N/A	N/A	0.02g^2/Hz (Figure 2)	
Vibration Gunfire–g^2/Hz Freq range (Hz)	N/A								MIL–STD–810B, Meth 519.1 RP: 0.02 g^2/Hz SDC: 0.026 g^2/Hz HE: 0.038 g^2/Hz (Figure 2)	
Acceleration Shock (S) Load (L) G (PK), time Waveform	Transportation (in container): 25 g, 25 ms, 1/2 sine wave Loading/unloading (in container): IAW MIL–STD–810B, Method 516, Procedure II								Shock 15 g 11 ms 1/2 sine wave	Load Vert: +7 to −1g Long: ±1.5g Lat: ±1.1g
Acoustic vib, dB	140 dB 53 – 6,784 Hz									
Humidity, lb water per lb dry air	100% humidity over a temperature range from 0°C (0.004 lb water per lb dry air) to 35°C (0.031 lb water per lb dry air)									
Water–rain(R) immersion(I) rate/time	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Corrosion potential salt spray test	N/A	N/A	Severe	N/A	N/A	Severe	N/A	Severe	Severe	

NEG = NEGLIGIBLE N/A = NOT APPLICABLE

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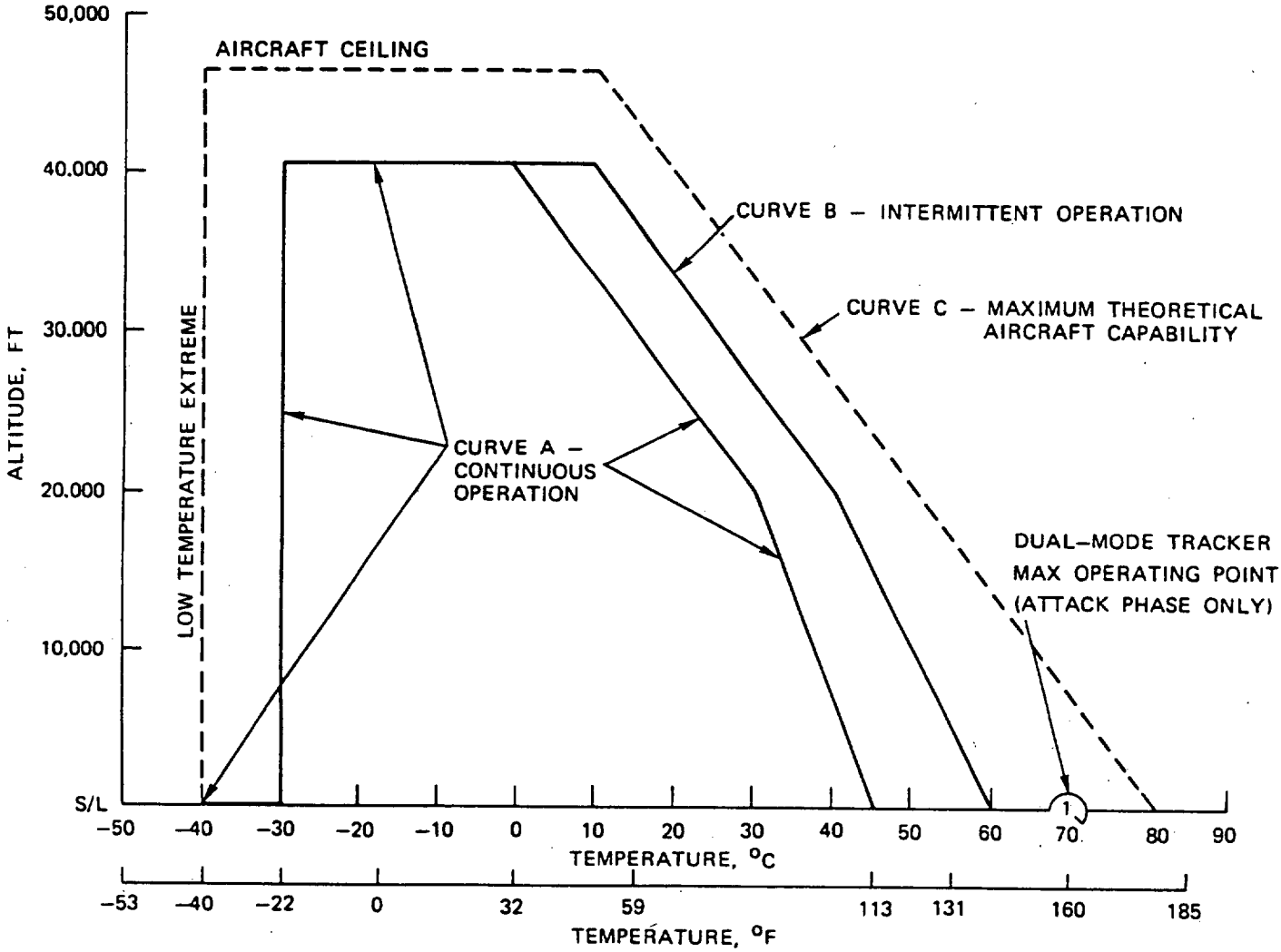


Figure 1. Tracker Operating Temperature/Altitude.

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Curve C shall be substituted for the intermittent conditions of Curve B and Point 1.

d. The tracker shall be capable of starting at a low temperature condition of -40°C and perform as specified in 3.4 and 3.5 after the warm-up period specified in 3.3.10.

3.3.9.3 Humidity. The tracker shall withstand the effects of humidity in accordance with MIL-E-5400.

3.3.9.4 Vibration. The tracker shall not suffer damage and shall meet the performance requirements of 3.4 and 3.5 when subjected to random vibration as specified in Figure 2.

3.3.9.5 Shock. The tracker shall not suffer damage or subsequently fail to provide the performance specified in 3.4 and 3.5 when subjected to the shock specified in Table I.

3.3.9.6 Gunfire. The tracker shall be capable of withstanding, in the operating mode, the vibrations induced by gunfire, as defined in Table I, with no physical damage or degradation of performance.

3.3.9.7 Acoustic Noise. The tracker shall be capable of performing as specified in 3.4 and 3.5 while being subjected to sound pressure levels specified in Table I and shall suffer no physical damage as a result of exposure to acoustic noise.

3.3.9.8 Salt Atmosphere. The tracker shall withstand exposure to salt-sea atmosphere as specified in MIL-E-5400.

3.3.9.9 Sand and Dust. The tracker shall withstand exposure to sand and dust as specified in MIL-E-5400.

3.3.9.10 Explosive Condition. The tracker shall not cause ignition of an ambient-explosive-gaseous mixture with air when operating in such an atmosphere.

3.3.9.11 Fungus Test. The tracker shall withstand exposure to fungus as specified in MIL-E-5400.

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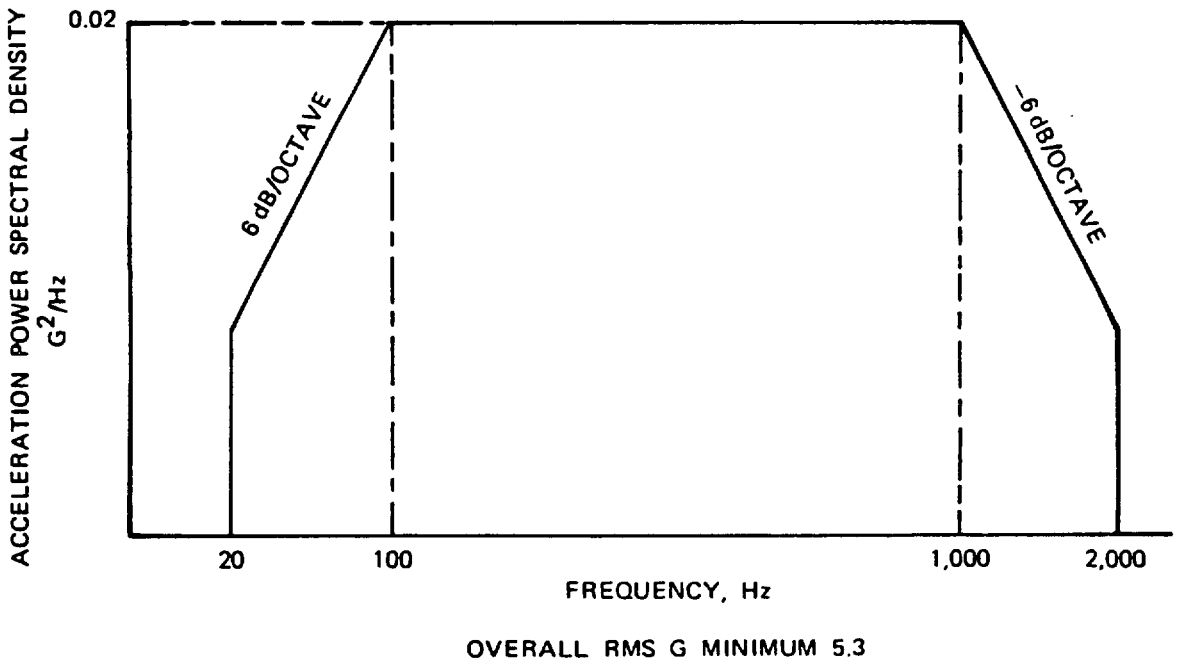


FIGURE 2. Random Vibration Levels for Tracker.

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3.3.10 Warm-up Time. The time required for the tracker to warm up to a temperature that allows normal operation shall not exceed 1 minute for temperatures above 0°C. An additonal warm-up period of 2 minutes shall be allowed for each 5.0°C ambient temperature decrement below 0°C after stabilization at any temperature in the range from 0°C to-40°C. Maximum allowable warm-up period, after stabilization at worst case low temperature condition of -40°C, shall not exceed 17 minutes.

3.3.10.1 Degraded Performance. Tracker degraded tracking perfomance will be permitted when extreme low temperature service conditions dictate operational capability prior to the warm-up period specified above. As a minimum, the tracker shall be capable of tracking nominal TV and LST targets specified in MIL-T-85115 within 5 minutes after being subjected to a cold-soak temperature environment of -26°C.

3.3.11 Input Electrical Power. The electrical power required by the tracker shall be as specified in the following subparagraphs.

3.3.11.1 Operating Power. The tracker shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

a. AC power (Three Phase, Y-Connected), 115/200 volts (V), 400 Hertz (Hz), Category B, 1100 volt-amperes (VA) total.

b. DC power, 28 V, Category B, 25 watts.

3.3.11.2 Degraded Performance. Tracker degraded performance shall be permitted for all voltage transients from 50 microseconds to 0.5 second in duration, at input voltage ranges between 50 and 160 volts alternating current (VAC) as shown in MIL-STD-704 (curves designated 2 and 3). Performance shall not be affected by short pulses of less than 50 microseconds. Tracker operation shall return to normal when the voltage transient expires. During degraded performance, the logic status shall not change due to the transient condition.

3.3.12 Cooling. The tracker shall require no special cooling other than ambient air in the installation environment.

3.3.12.1 Thermal Overload. The tracker shall provide thermal overload protection and shall automatically shut down when the temperature of the vidicon exceeds +85°C or when the temperature of the gimbal torquers exceeds +115°C.

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3.3.12.2 Installed Operating Temperature Checkpoint. Each production unit shall contain temperature sensing devices for monitoring the temperature of the most thermally critical components in accordance with existing design. The accuracy of the temperature sensing devices shall be within +1.7°C (3°F) of the true temperature and the location of the temperature sensing devices shall be approved by the procuring activity. The device outputs shall be capable of being monitored at the external TM connector (see 3.3.3.1).

3.3.13 Interface Definition. Interface definitions shall be as specified in 3.4.5

3.4 Performance. Unless otherwise specified in the contract, values set forth to establish the requirements of satisfactory performance apply to performance under standard conditions of 3.3.8 and service conditions of 3.3.9 and input power conditions of 3.3.11.

3.4.1 General Description. The Dual Mode Tracker (DMT) provides capability for acquiring and tracking either visually sighted or laser designated targets, and supplying electrical signals to associated avionics equipment to be used in computing weapons release information. The DMT consists of the following: a) Receiver-Processor Unit (RP), b) Signal Data Converter (SDC), and c) Heat Exchanger (HE).

The DMT configurations for A-4M and AV-8B aircraft are different, and are controlled by the following drawings:

Aircraft	Drawing Number
A-4M	3366020-100
AV-8B	3366020-101

The tracker shall enhance first-pass target acquisition of laser designated targets and the delivery of laser-guided missiles and bombs using the laser spot tracker functional capability. Either the laser or television mode of operation may be used to measure the aircraft-to-target line-of-sight (LOS) angles and inertial angular rates needed to solve the angular rate weapon delivery equations in the Weapons Delivery Computer (WDC) or Mission Computer (MC). The gimbal platform shall provide isolation from the airframe, and the angular freedom required to track ground targets during aircraft maneuvers.

3.4.1.1 Receiver-Processor. The Receiver-Processor shall consist of an inertially stabilized sensor element contained within a gimbal structure which stabilizes the sensor element and isolates it from aircraft motion. The gimbal order shall be azimuth, elevation and roll from the innermost sensor element to the mounting base. The Receiver-Processor shall contain, as a minimum, the dual mode optics assembly,

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the laser and TV sensors, the LOS rate sensors, LST analog processor, TV camera circuits, BIT source and driver circuits, and gyro cage electronics. The optical system shall provide the capability for laser and TV tracking. The laser and TV sensors shall include the signal preamplification as required. Rate integrating gyros (RIG) shall be used for stabilization and rate measurement in azimuth and elevation axes. A roll rate sensing gyro shall be mounted on the roll gimbal for stabilization damping of the roll servo. A torquer shall be used to drive each gimbal about its axis to meet the acceleration and rate requirements of this specification. Angle pick-offs shall be used to provide gimbal angle information about each axis. The Receiver-Processor units for A-4M and AV-8B configurations shall be identical.

3.4.1.2 Signal Data Converter. The Signal Data Converter shall contain all circuitry required to meet the tracker requirements of this specification except the circuitry that is contained in the Receiver-Processor. The Signal Data Converter shall contain digital signal conversion electronics for the tracker-to-WDC (A-4M), or tracker-to-MC (AV-8B), interface as appropriate for the specific configuration. The A-4M and AV-8B signal data converter configuration are identical, except for two SRAs: The A6 and A8 printed wiring boards. The part numbers for the A6 and A8 SRAs are given below for both A-4M and AV-8B configurations:

	Part Numbers	
	A6	A8
A-4M	3490650	3490670
AV-8B	3366200	3366216

All SRAs, other than A6 and A8, will be interchangeable between the two SDC configurations.

3.4.1.3 Heat Exchanger. The Heat Exchanger shall contain an electric heater, control electronics, fans, and sufficient heat exchange area to either cool or heat dry air gas to maintain the internal temperature of the Receiver-Processor within operating limits. The Heat Exchanger shall provide the transfer of heat into and out of the Receiver-Processor by means of a closed-loop cooling system using the aircraft nose compartment ambient air as the heat sink medium during the cooling process.

3.4.2 Tracker Operational Usage. The tracker shall have the capability to provide either manual or automatic search functions depending upon the operational mode selected by the pilot. The tracker shall, as a minimum, be capable of operating in the modes specified herein. The tracker shall provide automatic acquisition and tracking of laser designated targets when in the laser spot tracker mode of operation. The tracker shall provide lock-on and track capability of targets

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designated by the pilot when operating in the TV mode of operation. The tracker, when integrated with the aircraft avionics and ARBS equipment, shall provide accurate target LOS angle and inertial angular rate information to the WDC or MC (as appropriate for the aircraft).

3.4.2.1 Search. When operating in the search mode, the tracker shall function as specified herein.

3.4.2.1.1 LST Search. The tracker shall respond to WDC or MC controlled search options for laser target acquisition. The first, designated the LST NORMAL mode, is a single-bar roll and pitch stabilized azimuth search pattern. In this search mode the depression angle will be controlled by an input from the WDC or MC. This search mode is intended to provide target acquisition during the pre-attack phase of the flight to the target area. The minimum tracker gimbal limits for this mode shall be as specified in MIL-T-85115. The second, in the A-4M, designated the LST ARMAMENT DATUM LINE (ADL) mode, is a two-bar roll stabilized horizontal search pattern referenced to the aircraft ADL. This search mode in the AV-8B will provide a bow-tie scan pattern referenced to the aircraft. This mode is intended to provide the necessary search pattern for acquisition in a dive attack.

3.4.2.1.2 TV Search. The tracker shall provide for two target acquisition modes during TV operations. In the cage mode, the gimbal platform optical line of sight shall be electrically caged to input signals provided by the WDC or MC. These signals will be referenced to the aircraft longitudinal axis (ADL in A-4M). In the slew mode, the optical LOS shall be inertially stabilized and controlled upon command by the pilot with the cockpit controller via the WDC or MC.

3.4.2.2 Target Detection and Acquisition. Detection and acquisition of laser-designated targets within the optical field-of-view (FOV) shall be automatic. Acquisition of TV targets shall be accomplished by pilot manual operation.

3.4.2.3 Tracking Envelope. The tracker shall track visual and laser-designated targets within the operational envelope of the aircraft and provide outputs of gimbal angle and inertial angular rate as specified herein. The operational envelope of the aircraft is defined by the following parameters:

a.	Altitude:	0 to 45,000 feet MSL
b.	Velocity	250 to 550 knots indicated airspeed (KIAS)

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c.	Attitude:	Unrestricted (target LOS within tracker gimbal limits)
d.	Normal acceleration:	-1 to +5 gravity units(g)

The tracker shall impose no operational constraints upon the aircraft.

3.4.3 Target Characteristics. The tracker shall be capable of tracking targets typical of close support, interdiction and strike missions. Target types include, but are not limited to, troops, bunkers, tanks, trucks and convoys, parked aircraft, buildings, bridges, airfields, gun emplacements, SAM sites, and moving targets such as tanks, trucks, and ships. In general, the TV tracker shall be capable of acquiring and tracking any target that is detectable to the unaided eye. Targets may be either designated by laser or visually acquired.

3.4.3.1 Laser-Desginated Targets. The tracker shall be capable of acquiring and tracking laser-designated targets with the return energy characteristics specified in MIL-T-85115.

3.4.3.2 TV Targets. The TV tracker shall provide acquisition and tracking of the targets having the characteristic specified in MIL-T-85115.

3.4.4 Tracker Block Diagram. A block diagram of the tracker shall be as shown in Figure 3.

3.4.5 Interface Definitions. When integrated as part of the ARBS into the aircraft, the tracker shall conform to the interfaces specified in MIL-A-85055, and as specified in the following paragraphs.

3.4.5.1 Mechanical Interface. The tracker mechanical interface shall be in accordance with MIL-A-85055. The tracker equipment shall conform to the following installation control drawings:

a.	Dual Mode Tracker (A-4M) Dual Mode Tracker (AV-8B)	Drawing 3366020-100 Drawing 3366020-101
b.	Receiver-Processor (A-4M & AV-8B)	Drawing 3366021-500
c.	Signal Data Converter (A-4M & AV-8B)	Drawing 3366022-500

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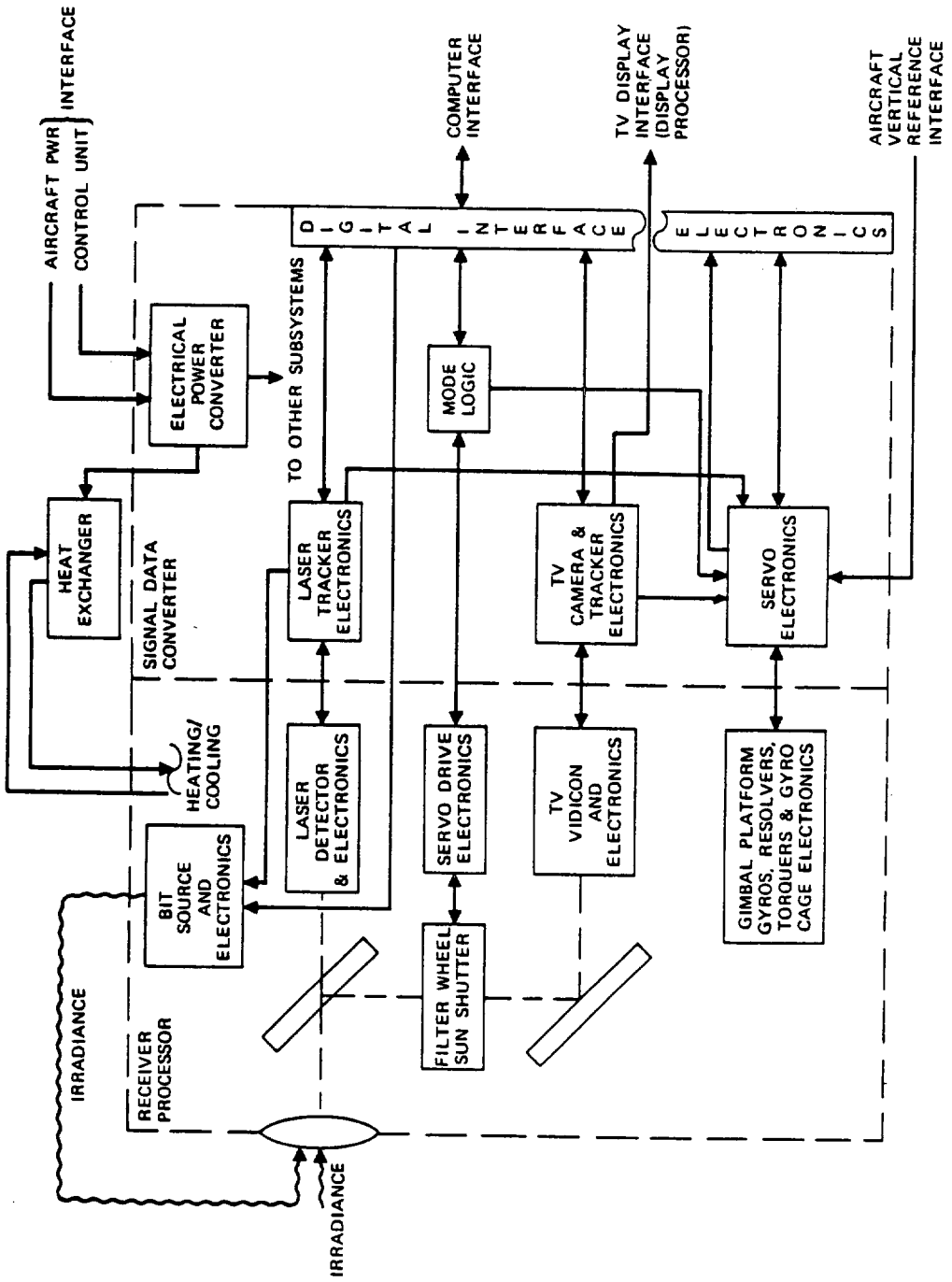


Figure 3. Dual Mode Tracker Block Diagram.

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d. Heat Exchanger (A-4M & AV-8B) — Drawing 3366030-500

e. Cooling/Heating Hose Assy (A-4M) — Drawing 3491400-500
Cooling/Heating Hose Assy (AV-8B) — 3366801-1 and 3366801-2

3.4.5.2 Environmental Interface. The tracker environmental interface shall be in accordance with MIL-A-85055.

3.4.5.3 Electrical Interface. The tracker shall interface electrically with the equipment specified below and in accordance with MIL-A-85055, the ARBS Electrical Interface Drawing 3366000-350 (A-4M) and the Interface Control Document 75J875900 (AV-8B).

3.4.5.3.1 A-4M Weapon Delivery Computer (WDC). For the A-4M aircraft, the tracker shall interface with the WDC, as shown in Figure 4 and Table II, and in accordance with MIL-A-85055. The input and output functions are defined in Tables II and III a.

3.4.5.3.1.1 WDC Logic Outputs. The tracker shall accept the following logic commands from the WDC:

a. Converter Interface Unit (CIU)-to-Tracker Discrete Word No. 1 (Refer to Table IIIa).

(1) Slew - Servo mode command

(2) Lock-on - TV tracker and servo mode command

(3) Correlate - LST tracker and servo mode command. Signal is true one-half computer cycle (20 milliseconds) after the tracker output correlate valid is true.

(4) Filter on -- This signal commands the tracker filter on and off. The timing is established in the computer software program.

(5) Roll reset - Servo mode command. Signal is true when commanded by the WDC and causes the tracker platform to return to the zero roll position, if tracker had previously been rolled more than 270 ± 45 degrees.

(6) Gate width - TV tracker command. The true state causes the TV gate size to be wide. The false state causes the gate to be narrow.

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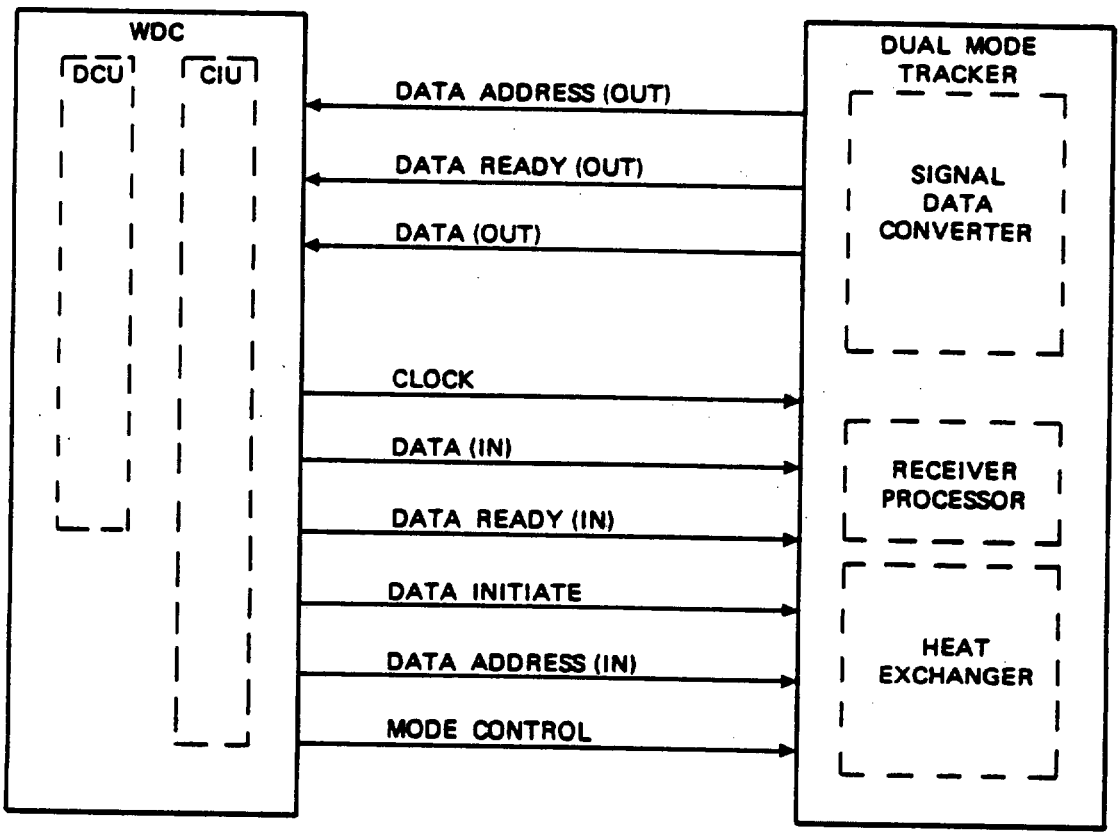


FIGURE 4. WDC/ARBS Dual Mode Tracker Interface.
(Applicable to A-4M Aircraft)

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TABLE II. WDC/Dual Mode Tracker I/O Interface.

Item No.	Signal Name	Characteristic
	WDC Input Signals From Dual Mode Tracker	
1	DATA READY (OUT) and DATA READY (OUT) COMPLEMENT	Bit 1 will be synchronous with the rising edge of CLOCK pulse.
2	DATA ADDRESS (OUT) and DATA ADDRESS (OUT) COMPLEMENT	Bit 1 will be synchronous with Bit 1 of DATA READY (OUT) and rising edge of CLOCK pulse, and have the following addresses: ADDRESS CODE Word / Bit: 15 16 17 18 Address 1: 0 0 1 0 Address 2: 1 0 1 0 Address 3: 0 1 1 0 Address 4: 1 1 1 0 Address 5: 0 0 0 1 Address 6: 1 0 0 1 Address 7: 0 1 0 1 Address 8: 1 1 0 1 Address 9: 0 0 1 1 Address 10: 1 0 1 1 Address 11: 0 1 1 1 Address 12: 1 1 1 1
3	DATA (OUT) AND DATA (OUT) COMPLEMENT	Bit 1 will be synchronous with Bit 1 of DATA READY (OUT) and DATA ADDRESS (OUT), and rising edge of CLOCK pulse. Date will contain twelve 20-Bit serial data words with the addresses as specified in item 2.
4	DATA READY (IN) and DATA READY (IN) COMPLEMENT	Bit 1 shall be synchronous with rising edge of CLOCK pulse.

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TABLE II. WDC/Dual Mode Tracker I/O Interface (Continued).

Item No.	Signal Name	Characteristic
	WDC Output Signals To The Dual Mode Tracker	
5	DATA ADDRESS (IN) and DATA ADDRESS (IN) COMPLEMENT	Bit 1 shall be synchronous with Bit 1 of DATA READY and rising edge of CLOCK pulse, and have the following addresses: ADDRESS CODE Word / Bit: 15 16 17 18 Address 1: 1 0 0 0 Address 2: 0 1 0 0 Address 3: 1 1 0 0 Address 4: 0 0 0 0
6	DATA (IN) and DATA (IN) COMPLEMENT	Bit 1 shall be synchronous with Bit 1 of DATA READY and DATA ADDRESS, and rising edge of CLOCK pulse. DATA shall consist of four 20-Bit serial words with addresses as specified in item 5.
7	CLOCK and CLOCK COMPLEMENT	50 kHz timing clock for all digital signals.
8	DATA INITIATE and DATA INITIATE COMPLEMENT	A high signal requests that all data words be transmitted by the subsystem to WDC.

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TABLE IIIa. Tracker Data Scaling

Data Word	Data Word Scaling			Polarity/Remarks
	± F. S.	#BITS	RANGE	
A. Data From Tracker				
1. Discrete Word No. 1	NA	NA	NA	NA
2. Azimuth Resolver	± 10 VDC	14 (1)	± 180°	+ = Gimbal is Right
3. Elevation Resolver	± 10 VDC	14 (1)	± 180°	+ = Gimbal is Up
4. Roll Resolver	± 10 VDC	14 (1)	± 180°	+ = CCW (Left Wing Down)
5. Azimuth Rate Error	± 10 VDC	12	± 31.25 mr/sec	+ = Gimbal Rate is Right
6. Elevation Rate Error	± 10 VDC	12	± 31.25 mr/sec	+ = Gimbal Rate is Up
7. Elevation TV Tracker Error	± 10 VDC	12	± 10.0 mr	+ = Target is Up
8. Azimuth TV Tracker Error	± 10 VDC	12	± 10.0 mr	+ = Target is Right
9. Azimuth TV Tracker Rate	± 10 VDC	12	± 50 mr/sec	Growth
10. Elevation TV Tracker Rate	± 10 VDC	12	± 50 mr/sec	Growth
11. LST Azimuth Error	± 10 VDC	12	± 3.33 mr	+ = Target is Right
12. LST Elevation Error	± 10 VDC	12	± 3.33 mr	+ = Target is Up
B. Data to Tracker				
1. Discrete Word No. 1	NA	NA	NA	NA
2. Discrete Word No. 2	NA	NA	NA	LST Code
3. Azimuth Command	± 10 VDC	12 (1)	(2)	+ = Gimbal Go Right
4. Elevation Command	± 10 VDC	12 (1)	(2)	+ = Gimbal Go Up

(1) Includes sign bit.

(2) Position Command Mode = Sine of gimbal angle
Rate Command Mode = ± 500 mr/s if Rate Scaling Bit = 1
= ± 125 mr/s if Rate Scaling Bit = 0

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(7) FIT pass - Signal is true if the WDC determines that the tracker response is proper for WDC monitored FIT tests.

(8) Pull-up warning - Signal is false if the WDC determines that aircraft ground clearance or blast avoidance pull-up range has been reached.

(9) Bit source on - This signal is true as commanded by the WDC during BIT and FIT test mode step #6. The signal shall cause the LST/TV BIT target source in the tracker to be turned on or activates LST guidance circuitry during FIT.

(10) Yellow filter select - This signal is true for yellow filter and false for red filter

(11) Linear Track Command - The WDC determines that the signal strength is adequate to begin linear tracking (LST Mode), and commands that action.

b. CIU-to-tracker Discrete Word No. 2 (see Table IIIb and the Appendix) LST code. The LST code shall be input to the tracker in octal format using Data Bits 7 through 15 of the CIU-to-tracker Discrete Word No. 2. Bits 7 through 9 shall contain the fourth digit of the octal code with Bit 9 being the MSB. Bits 10 through 12 shall contain the third digit of the octal code with Bit 12 being the MSB. Bits 13 through 15 shall contain the second digit of the octal code with Bit 15 being the MSB.

c. CIU-to-tracker Mode Control Lines (see Figure 4). These control lines shall be open or grounded by the CIU (MIL-W-85057) to control the different tracker operating modes (see Table V).

3.4.5.3.2 AV-8B Mission Computer (MC) Interface. For the AV-8B aircraft, the tracker shall interface with the MC, as shown in Figure 5, and in accordance with MDC A5232. The input and output functions are defined in the Appendix.

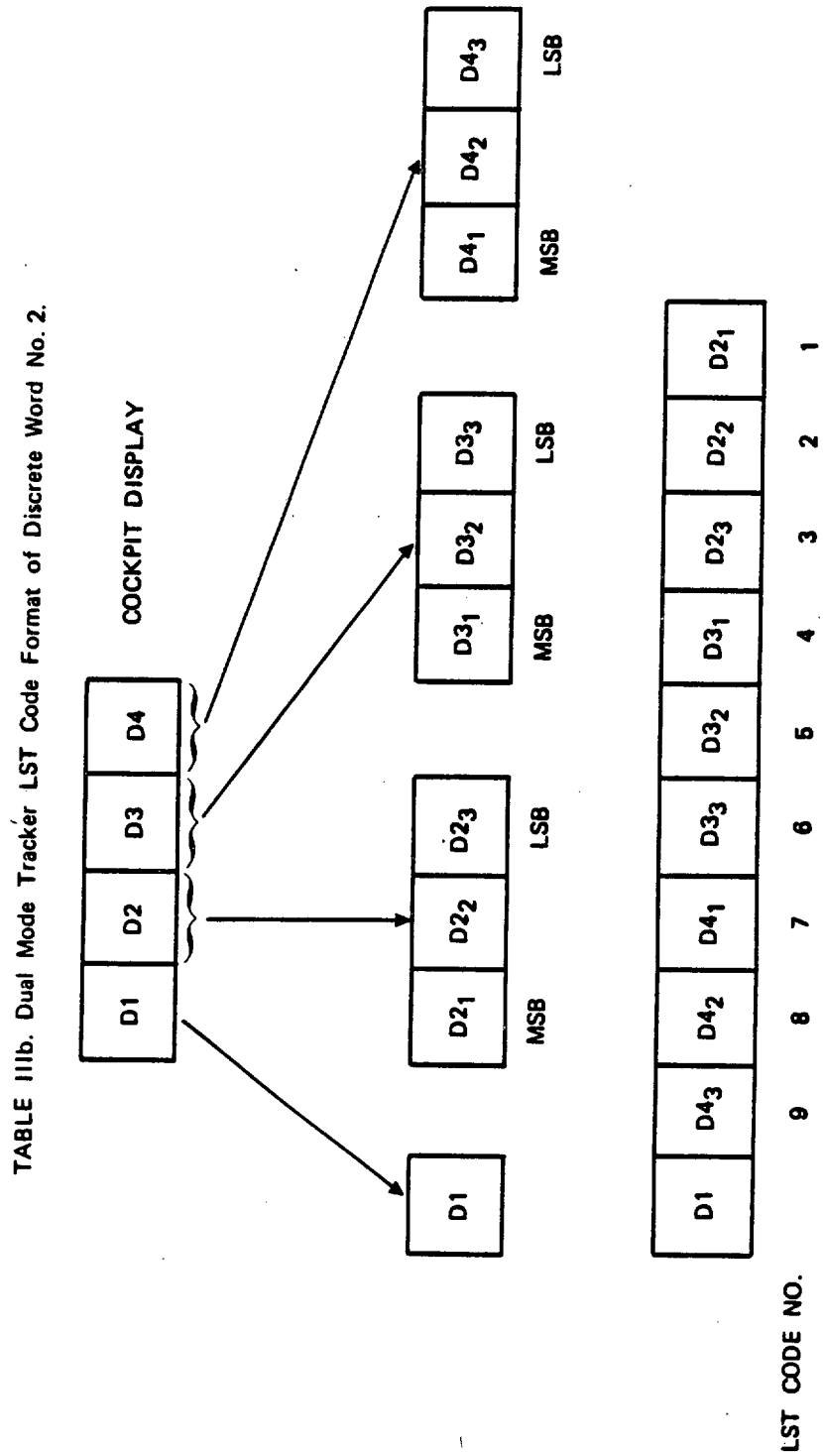
3.4.5.3.2.1 MC Logic Outputs. The tracker shall accept the following logic commands from the MC:

a. MC to DMT Discrete Word No. 1 (refer to the Appendix).

(1) Slew - Servo mode command.

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TABLE IIIb. Dual Mode Tracker LST Code Format of Discrete Word No. 2.



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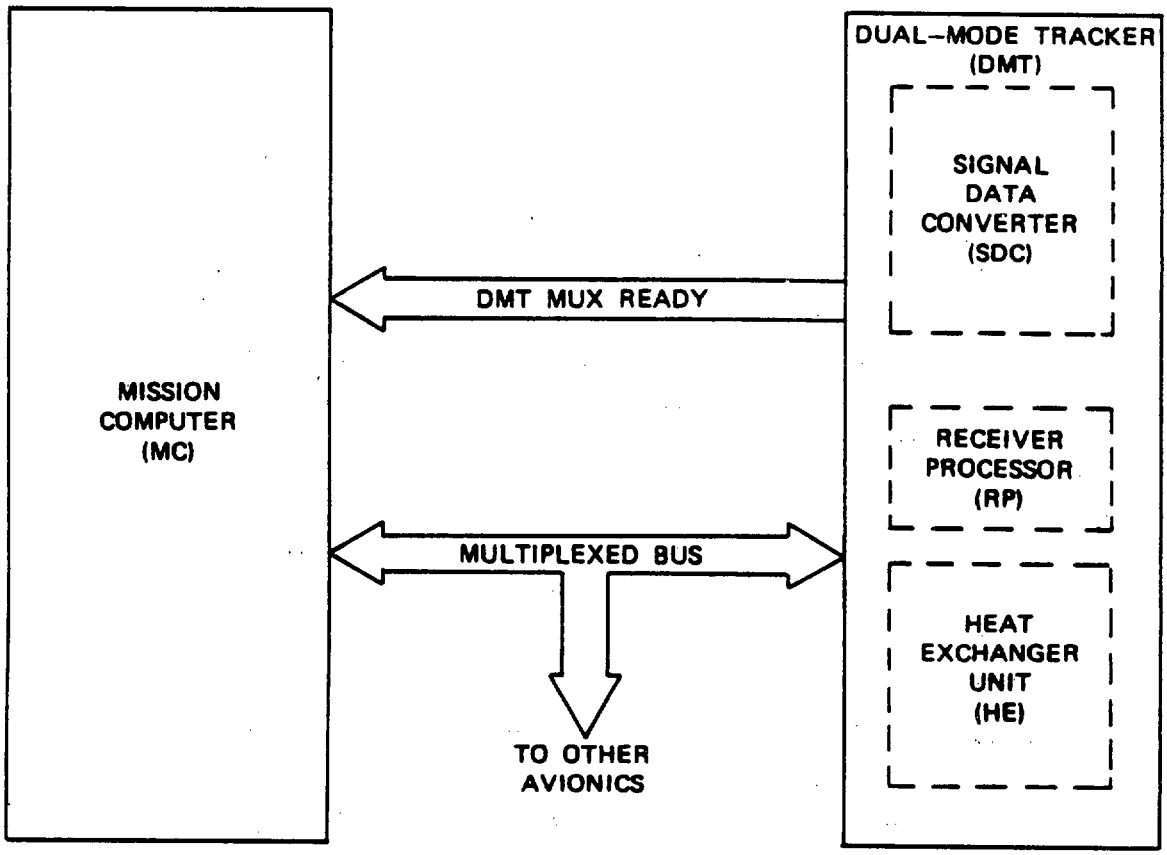


FIGURE 5. MC/ARBS Dual Mode Tracker Interface.
(Applicable to AV-8B Aircraft)

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(2) Lock-on - TV tracker and servo mode command.

(3) Correlate - LST tracker and servo mode command. Signal is true one-half computer cycle (25 milliseconds (ms)) after the tracker output correlate valid is true.

(4) Filter on - This signal commands the tracker filter on and off. The timing is established in the computer software program.

(5) Roll reset - Servo mode command. Signal is true when commanded by the MC and causes the tracker platform to return to the zero roll position, if tracker had previously been rolled more than 270 ± 45 degrees.

(6) Gate width - TV tracker command. The true state causes the TV gate size to be wide. The false state causes the gate to be narrow.

(7) FIT pass - Signal is true if MC determines that tracker repsonse is proper for MC monitored FIT tests. (Used in ground maintenance).

(8) Pull-up warning - Signal is false if MC determines that aircraft ground clearance or blast avoidance pull-up range has been reached. Will cause break-X symbol on TV display.

(9) Bit source on - This signal is true as commanded by the MC during BIT and FIT test modes step #6. The signal shall cause the LST/TV BIT target source in the tracker to be turned on or activates LST guidance circuitry during FIT.

(10) Yellow filter select - This signal is true for yellow filter and false for red filter.

(11) Mode control bits - These three bits are used by the MC to control the operating mode of the DMT.

(12) Linear track command - The MC determines that the signal strength is adequate to begin linear tracking (LST Mode), and commands that action.

b. MC-to-tracker Discrete Word No. 2 (see Table IIIb and the Appendix) LST code. The LST code shall be input to the tracker in octal format using Bits 1 through 9 of the MC-to-tracker Discrete Word No. 2. Bits 1 through 3 shall contain the fourth digit of the octal code with Bit

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3 being the MSB. Bits 4 through 6 shall contain the third digit of the octal code with Bit 6 being the MSB. Bits 7 through 9 shall contain the second digit of the octal code with Bit 9 being the MSB.

3.4.5.3.3 DMT Logic Outputs. The tracker shall provide the following logic status signals to the WDC or MC:

a. DMT Discrete Word No. 1.

(1) FIT A -

(2) FIT B - During Fit these signals shall be true as a function of the SDC switch postion (a sample system implementation is shown in Table IV for information only).

(3) FIT C -

(4) Linear track - This signal shall be true when the azimuth and elevation crossings have occurred, the LST AGC has settled, correlate valid is true, and linear track command is received from the WDC or MC.

(5) Correlation valid - This signal shall be true when the correlation requirements specified herein are satisfied.

(6) Pulse valid - After correlation valid is true, this signal shall be true whenever a valid (selected code) laser signal exceeds threshold.

(7) Tracker ready - This signal is true when the DMT is ready for operational use. The signal is false during roll reset and when the temperature limits are exceeded.

(8) Rolled over - This signal shall be true when the absolute value of the roll gimbal angle exceeds 270 ±45 degrees or is in the process of resetting.

(9) Overheat/turn-off - This signal shall be true whenever the temperature at the SDC temperature sensor exceeds the overheat limits, or the temperature at the RP temperature sensor exceeds the overheat limits. This signal shall cause tracker power shutdown to occur within 130 ms ±40 percent. (Refer to 3.3.12.1 for details of overheat protection.)

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TABLE IV. FIT Mode Logic.

FIT POS	FIT A B C	WDC OR MC OUTPUTS							SERVO MODES			
		SLEW	LOCK ON	CORR	FILT ON	WIDE GATE	FIT PASS	BIT SOURCE ON	POS	RATE	TV TRK	LST TRK
0	1 1 1	X	X	X	X	X	X	X	X	X	X	X
1	1 1 1	X	X	X	X	X	X	X	X	X	X	X
2	1 1 1	X	X	X	X	X	1	X	X	X	X	X
3	1 1 0	X	X	X	X	X	1	X	0	0	0	0
4	1 0 1	X	1	0	0	0	1	0	0	0	1	0
5	1 0 0	X	1	0	0	0	1	X	0	0	1	0
6	0 1 1	X	0	1	1	X	1	1	0	1	0	1
7	0 1 0	X	X	X	X	X	1	X	1	0	0	0
8	0 0 1	X	X	X	X	X	1	X	1	0	0	0
9	0 0 0	X	X	X	X	X	1	X	0	0	0	0

1 = True
0 = False
X = Don't Care

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(10) Parity - This signal shall be true if the parity of the fourth (last) word in the A-4M CIU-to-tracker serial data frame is correct. This discrete is a spare in the AV-8B.

(11) Not in FIT - This signal shall be true when the SDC FIT cable (wrap-around) is not connected.

(12) Underheat - This signal shall be false when the RP temperature sensor is colder than the underheat limits.

(13) Sun shutter monitor - This signal shall be true when the vidicon sun shutter is closed or closing.

(14) Good signal - This signal shall be true whenever valid (selected code) laser signal is of sufficient strength to sustain LST linear track.

3.4.5.3.4 Cockpit Control Interface. The tracker shall accept operating control power from the ARBS control subsystem.

3.4.5.3.5 TV Display Interface. The tracker shall provide properly matched video signals to the aircraft TV monitor (IP-936/AXQ) for the A-4M, and the Processor/Multipurpose Display for the AV-8B. The tracker shall perform as specified herein when operating into a 91 ohm load, and shall not be damaged when operating into no-load (open circuit).

3.4.5.3.6 Vertical Reference Interface. The tracker shall accept a vertical gyro reference synchro signal from the aircraft for stabilization of the roll axis in accordance with 3366000-350 (A-4M) or ICD 75J875900 (AV-8B).

3.4.5.3.7 Instrumentation Interface. Electrical signal functions necessary for use in a maintenance or for in-flight telemetry recording shall be brought out to a test connector. These shall include those functions listed in the Electrical Interface Data Drawing 3366000-350. All functions shall be electrically isolated so that shorting between any function and ground shall not damage the tracker or impair the tracker operation, except where approved by the procuring activity. The connector shall be mounted in accordance with Drawing 3366022-500.

3.4.6 Tracker Characteristics. Tracker characteristics shall be as specified in the following subparagraphs.

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3.4.6.1 Tracker Performance. The tracker shall meet the following performance requirements under the service conditions specified in 3.3.9.

3.4.6.1.1 Operating Modes. The tracker shall operate in the following modes selectable by the pilot.

3.4.6.1.1.1 OFF. In the OFF mode the tracker shall have all power removed. The vidicon shall not be damaged if the seeker is directed at the sun in the OFF mode. When any operating mode is selected from the OFF mode, the tracker shall automaticlly sequence through STANDBY for a time period not to exceed the warm-up time specified in 3.3.10.

3.4.6.1.1.2 STANDBY. In the STANDBY mode, the tracker shall provide power to the vidicon, gyro spin motors, heaters, and fans if temperature requires, and STANDBY electronics. The vidicon shall be blanked. At a time not to exceed 15 seconds after entering the STANDBY mode, the tracker shall be electrically slaved in roll to the aircraft vertical reference.

3.4.1.1.3 TV. Upon selection of the TV mode of operation, the tracker shall be capable of tracking the specified TV targets with appropriate rate-aided commands from the WDC or the MC.

3.4.6.1.1.4 LST. Upon selection of the LST mode of operation, the tracker shall be capable of tracking the specified laser targets with appropriate rate-aided commands from the WDC or the MC.

3.4.6.1.1.5 Boresight (BST). Upon selection of the BST mode, the tracker shall be roll stabilized with the LOS electrically caged to azimuth and elevation gimbal positions as commanded by the WDC or the MC.

3.4.6.1.1.6 Built-In-Test (BIT). Upon selection of the BIT mode, the WDC or MC initiates and controls an operational readiness test of the tracker. The tracker shall provide a suitable built-in-test source to verify operational readiness in the TV and LST modes.

3.4.6.1.2 Receiver-Processor Control. The tracker shall include provisions for WDC or MC control of either the position or rate of the tracker optical LOS. Polarity of command and output signals shall be as specified in Table IIIa. A summary of command signals and control mode responses is given in Tables Va and Vb and the Appendix.

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TABLE Va. Summary of Tracker Operating and Control Modes (Azimuth and Elevation Servo).

Input Mode Signals	DMT On	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Mode A	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0
	Mode B	1	1	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0
	Mode C	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0
	Slew				0	1	0	1	0	1	0	1	0	1	0	1	See LST	See LST
	Lock On				0	0	1	0	0	0	1	0	0	0	1	1		
	Correlate								0	0	0	0	1	1	1	1		
	Filter On (1)				0	0	1	1	0	0	1	1	(9)	(9)	1	1		
	Gate Width (1)				1	1	0	0	1	1	0	0	0	0	0	0		
	AZ Command (1)			(5)	(5)	(6)	(7)	(8)	(6)	(6)	(7)	(8)	(7)	(8)	(7)	(8)		
	EL Command (1)			(5)	(5)	(6)	(7)	(8)	(6)	(6)	(7)	(8)	(7)	(8)	(7)	(8)		
Servo Response Modes	Position Control	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	See LST	See LST
	Rate Control	0	0	0	0	1	1	1	0	1	1	1	1	1	1	1		
	TV Track	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0		
	LST Track	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0		
System Mode		OFF (4)	WDC (4)	STBY	TV				LST								BST	BIT

Notes:

(1) These input signals are not used to determine the servo mode. They are included in this table for information and understanding of total ARBS system operation.
(2) Deleted.
(3) AZ and EL torquers are disabled in STBY if RP temperature is above 10°F (underheat invalid).
(4) These modes will not exist in ARBS operations but can occur during tracker testing.
(5) Command from WDC or MC shall be used as AZ and EL position command to point RP optical axis.
(6) Commands from WDC or MC shall be used as AZ and EL rate commands to point RP optical axis.
(7) Commands from WDC or MC shall be used as AZ and EL tracking rate aiding commands.
(8) Commands from WDC or MC shall be used as AZ and EL rate commands to keep RP pointed at same ground position.
(9) This signal is delayed by the WDC or MC until after linear track goes true.

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TABLE Vb. Summary of Tracker Operating Modes (Roll Servo).

| Input Mode Signals | DMT On
Mode A
Mode B
Mode C
Roll Reset | 1 1 1

3.4.6.1.2.1 Platform Gimbal Position. The tracker shall provide outputs of azimuth, elevation and roll gimbal positions. The outputs shall indicate azimuth, elevation and roll gimbal angles with respect to the Receiver-Processor zero gimbal reference position.

3.4.6.1.2.2 Roll Vertical Reference. The tracker shall align the platform roll axis to the aircraft vertical reference synchro input in STANDBY, TV, BST, and LST modes.

3.4.6.1.3 Target Tracking. The tracker shall be capable of acquiring and tracking the targets with characteristics specified by 3.4.3 and shall provide the tracking performance specified in the following subparagraphs.

3.4.6.1.3.1 Tracking rates. The gimbal platform shall respond to azimuth and elevation angular rate commands equal to the sum of an aided tracking rate generated by the WDC or MC and a rate feedback proportional to the tracking position error generated by the TV or LST tracker. The range of values of the aided tracking rate shall be as specfied by MIL-T-85115. The polarity is defined in Table IIIa.

3.4.6.1.3.2 Tracking Rate Accuracy. The tracking rate accuracy as measured at the output of the tracker on azimuth and elevation channels shall be as specified by MIL-T-85115.

3.4.6.1.3.3 Tracker Rate Noise. The root-mean-square (rms) value of the noise of the tracker angular rate error output shall be as specified in MIL-T-85115. The specified tracking noise shall not be exceeded during testing when the tracker is tracking specified targets and the angular rate error output is processed by a linear filter which has the following O/I transfer functions:

$$G(s) = \frac{10.75s}{(10s + 1)(0.25s + 1)}$$

3.4.6.1.4 Slew Rates. The tracker shall be capable of slewing the optical LOS at high inertial rates as specified by MIL-T-85115.

3.4.7 LST Requirements. The LST shall perform as specified herein.

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3.4.7.1 Laser Sensitivity. Laser sensitivity shall be as specified in MIL-T-85115.

3.4.7.2 LST Field of View. The LST acquistion field of view shall be as specified in MIL-T-85115.

3.4.7.3 LST Off-Axis Target Sensitivity. The LST shall not detect or process target return signals of up to 1,000 times the threshold sensitivity level that are located outside the normal field of view defined in 3.4.7.2. An acceptable alternative is to detect and process the target return signals, provided that the LST tracks the target after correlation, as specified in 3.4.6.1.3.

3.4.7.4 Laser Signal Processing. The laser signal processor shall provide azimuth and elevation tracking signals. The circuit shall receive low level signals from the sensor element and generate tracking commands for the servo electronics.

3.4.7.4.1 Background Illumination. The requirements of 3.4.7 shall apply with an ambient solar background illumination of 6.0 microwatts per square centimeter per Angstrom, within the spectral bandwidth of the receiver on a diffuse background whose reflectance is 50 percent.

3.4.7.4.2 False Acquistion Rate. The false acquisition rate shall be less than two per hour, with a solar background up to and including the level specified in 3.4.7.4.1.

3.4.7.4.3 Dynamic Range. The LST shall have a minimum tracking dynamic range of 100,000 to 1 above the target return signal level specified in 3.4.7.1.

3.4.7.4.4 Missed Command (Tracking). There shall be no more than one missed tracking command during any 1-minute period when the LST is tracking a source (3.4.3.1) whose incident target return energy is between 4:1 and 100,000:1 of the energy levels specified in 3.4.7.1.

3.4.7.4.5 Correlation. The LST shall provide a discrete signal to the WDC or MC denoting correlation after the valid target return pulses (above threshold) are received at the proper pulse spacing in accordance with MIL-T-85115. The LST shall commence tracking upon receipt of a discrete command from the WDC or MC. The LST shall remain in the track mode until the discrete command is removed by the WDC or MC.

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3.4.7.4.6 Decorrelation. The LST shall automatically switch the correlation discrete to a false condition after missing pulses as specified by MIL-T-85115.

3.4.7.4.7 Multiple Target Return Discrimination. Discrimination of multiple targets shall be performed in accordance with MIL-T-85115.

3.4.7.4.8 Pulse-to-Pulse Response. The LST shall process a sequence of target return pulses with a maximum amplitude variation of 8:1 and operate to the requirements herein.

3.4.7.4.9 Target Return Overload. The LST shall recover from a target return overload within 1.5 second when subjected to a target return of up to 500 times the energy level specified in MIL-T-85115.

3.4.7.4.10 Sun Blanking. The LST shall not lock on or false acquire when either specular reflected (sun glint) or direct sunlight is within or near the edge of the LST field of view. During this condition, the tracker is not required to track a laser target.

3.4.7.4.11 Noise Variable Compensator. The LST shall automatically compensate for changing total system noise over a total dynamic noise range from minimum (low temperature, no background) to worst case (high temperature, maximum background) to provide the maximum possible LST sensitivity, while maintaining compliance with the specified requirements for false alarm rate, dynamic range, pulse-to-pulse response and multiple target return discrimination.

3.4.7.4.12 LST Preamplifier Protection. The LST preamplifier shall not sustain damage when a laser designator is directed at the seeker and is within the LST field of view. Preamplifier circuit protection design shall be as approved by the procuring activity.

3.4.8 TV Requirements. The TV shall conform to the requirements specified herein.

3.4.8.1 TV Light Level Range. The TV camera shall provide video for display and tracking with scene brightness levels of 20 to 10,000 footlamberts as measured at the scene.

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3.4.8.2 Target Contrast. The TV camera and tracker shall lock on and track targets having contrast relative to their backgrounds as specified in MIL-T-85115.

3.4.8.3 Vidicon Sun Protection. The TV vidicon shall be protected from sun damage in the event that the seeker is directed at the sun.

3.4.8.4 TV Flare Light Sensitivity. The TV performance specified herein shall not be degraded by the presence of the sun or other bright source at any angle greater than 10 degrees off the optical axis of the tracker.

3.4.8.5 TV Gate Width. The TV acquisition gate width shall be as specified by MIL-T-85115.

3.4.8.6 TV Field of View. The TV field of view shall be as specified in MIL-T-85115.

3.4.8.7 TV Signal Processing. The TV camera and TV tracking circuit designs shall provide azimuth and elevation tracking error signals. The signal processor shall receive a low level signal from the vidicon preamplifier and generate tracking error signals for the servo electronics and video for TV displays.

3.4.8.7.1 Amplifier. The video amplifier shall consist of two parts. The video from the preamplifier shall be amplified and sent to the TV tracker video processor. On a separate path, the provisions shall be made for video output for one TV monitor display and one instrumentation output. This includes amplification, key clamping, the addition of sync and tracker crosshair signals and output buffering for the TV display and instrumentation signal. The bandwidth of the amplifier shall be consistent with the bandwidth of the preamplifier. The voltage gain will be adjusted according to the command from a video level automatic gain control circuit. The input will be differentially received.

3.4.8.7.2 Camera Circuitry. The camera circuitry shall operate to provide proper drive and sweep voltages for the selected vidicon.

3.4.8.7.3 TV Raster. The nonlinearity of the picture shall not exceed 5 percent over the entire raster. The horizontal sweep shall be from left to right across the scene being viewed with the vertical sweep progressing from top to bottom across the scene. The interlace shall be postive 2 to 1 and the raster format ratio shall be 4 X 3 horizontal by vertical.

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3.4.8.7.4 Sweep Loss Circuit. A sweep loss circuit shall be included to blank the vidicon when proper deflection sweeps are not present.

3.4.8.7.5 TV Tracker Video Input. The TV camera circuit shall provide a signal from TV tracker signal processing.

3.4.8.8 TV Tracker Crosshair Generator. The TV tracking circuitry shall provide crosshair signals to the TV camera circuitry showing the position of the track point for display on the TV monitor.

3.4.8.9 Tracking Error Signals. The TV tracking signals shall be provided both as inputs to the gimbal servo electronics and as an output of the tracker. Two channels of signal are required: up-down (elevation) and left-right (azimuth) outputs shall be proportional to the angular position of the target with respect to the TV optical LOS.

3.4.8.10 Pull-up Warning. The pull-up warning discrete shall cause the tracker composite video signal to be altered so that a flashing "X" warning signal is displayed on the aircraft video monitor.

3.4.9 Boresight Accuracy. The tracker LOS shall be aligned to within 2 milliradians (mr) of the Receiver-Processor baseplate mounting surface when the WDC or MC azimuth and elevation position commands are both zero.

3.4.10 Cooling System Leak Rate. The overall leak rate of the closed-loop cooling system shall not exceed 40 standard cubic centimeters per hour at a pressure of one atmosphere. Provisions shall be made to prevent water vapor condensation on internal surfaces of the Receiver-Processor. These provisions shall not require any maintenance action more than once every three months under the humidity conditions as specified in Table I.

3.5 Detail Requirements. The tracker elements shall have the characteristics specified in the following subparagraphs.

3.5.1 Receiver-Processor. A three-axis gimbal structure shall provide inertial stabilization of the optical platform. The gimbal platform shall contain the dual mode optics, laser energy detector, TV vidicon, signal preamplifiers, rate integrating gyros, gimbal angle sensors, gimbal torquers, and electronics as required to conform to this specification.

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3.5.1.1 Gimbal Structure. The gimbal structure shall provide isolation from aircraft base motion about the azimuth, elevation, and roll axes.

3.5.1.1.1 Gimbal Elements. The innermost gimbal element shall be the optical platform containing the stabilized optics assembly, laser and TV sensors, signal preamplifiers, rate integrating gyros, and other mechanisms and electronics as required to meet the requirements of this specification. This element shall rotate about the azimuth axis with respect to the second gimbal. The second gimbal shall rotate about the elevation axis with respect to the roll gimbal. The roll gimbal shall provide roll axis freedom of the stabilized platform with respect to the mounting base. The roll axis shall be aligned with the aircraft attitude gyro reference.

3.5.1.1.2 Tracking Gimbal Freedom. The tracker shall have a minimum gimbal freedom as specified in MIL-T-85115.

3.5.1.1.3 Base Motion Disturbance/Response. The tracker shall meet the requirements of 3.4.2.3 when the tracker mounting base motions are as specified in MIL-T-85115.

3.5.1.1.4 Gimbal Mass Unbalance. After assembly, the azimuth and elevation gimbal structures shall be balanced to less than 0.2 ounce-inch of torque per "g" acceleration applied in a perpendicular plane to the gimbal rotational axis within the gimbal rotational limits. The roll gimbal shall be balanced to within 10 ounce-inches per "g" applied in a plane perpendicular to the roll axis.

3.5.1.1.5 Gimbal Alignment. The mechanical roll axis of the tracker is defined as a line that is orthogonal to the tracker mounting flange. The optical axis of the sensor optical element shall be aligned parallel to the roll axis of the tracker during factory adjustment of the elevation and azimuth gimbal pick-offs to electrical zero. The plane containing the azimuth and elevation axis shall rotate about the elevation axis. This plane shall be orthogonal to the roll axis of the tracker to within 2 mr during adjustment of the gimbal pick-off electrical zero. The azimuth and elevation rotation axes shall be mutually orthogonal to within 6 mr. The elevation rotational axis will be used to define the horizontal plane through the tracker roll axis during adjustment of the roll axis gimbal pick-offs.

3.5.1.1.6 Gimbal Stops. Gimbal stops shall be provided at gimbal limits of the azimuth, elevation and roll axes. They shall be con-

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structed to prevent damage to gimbal hardware assemblies during aircraft landing and take-off and when the platform is driven into the stops in any manner.

3.5.1.1.7 Gimbal Zeroing Reference Pins. The tracker gimbals shall be provided with adjustable zeroing pins for establishing the zero gimbal angle references for adjustments to the gimbal angle sensors.

3.5.1.2 Gimbal Angle Sensors. Single-speed resolver pick-offs having infinite resolution shall be mounted to each gimbal axis to provide sine and cosine measurements of their respective angles.

3.5.1.2.1 Azimuth Axis. The azimuth axis resolver pick-off shall provide measurements of azimuth angle between the angle limits with an accuracy of 2 mr of arc. The electrical zero of the pick-off shall be adjusted to within 1 mr of zero gimbal angle. Zero gimbal angle is defined by alignment of the optical axis parallel to the roll angle of the gimbal platform.

3.5.1.2.2 Elevation Axis. The elevation axis resolver shall provide measurement of elevation angle between the angle limits within an accuracy of 2 mr of arc. The electrical zero shall be adjusted to within 1 mr of zero gimbal angle defined by the alignment of the LOS optical axis parallel to the roll axis of the gimbal platform.

3.5.1.2.3 Roll Axis. Roll axis pick-off sensor(s) shall be provided as required to perform stabilization of the roll gimbal to local vertical and to provide a measurement of roll gimbal angle with respect to the aircraft. Roll stabilization shall be within an accuracy of 5 mr with respect to aircraft roll attitude as measured by the aircraft vertical reference system. The roll pick-off sensor(s) shall provide tracker roll angles with respect to the aircraft mounting base to an accuracy of 4 mr over roll angles of ±30 degrees. The tracker shall provide roll angle information accurate to 1 degree for roll angles of ±360 degrees.

3.5.1.3 Gimbal Torquers. Gimbal torquers shall be mounted coaxial to each rotation axis to provide gimbal drive power. The azimuth and elevation torquers shall provide full torque capability over the limits of gimbal travel. The azimuth torquer shall be capable of providing acceleration to the inner gimbal element of up to 2.5 radians per second per second. The elevation gimbal torquer shall be capable of accelerating the elevation gimbal and inner element to 1.5 radians per second per

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second. The roll axis torquer shall be capable of supplying torque to the roll gimbal to provide roll platform stabilization for roll rates up to 220 degrees per second and accelerations of 500 degrees per second per second.

3.5.1.4 Angular Rate Sensors. The rate integrating gyros shall be mounted to the inner platform element to provide LOS stabilization and rate feedback measurement of tracker LOS angular rate. A rate gyro shall be mounted to the roll gimbal to provide rate damping for servo stabilization of the roll gimbal.

3.5.1.4.1 Rate Integrating Gyro Parameter Specifications. The rate integrating gyros shall conform to the following requirements:

a. Random drift: less than 5 degrees per hour peak-to-peak.

b. G-insensitive drift: less than 12 degrees per hour.

c. G-sensitive drift: less than 10 degrees per hour per "g"

d. Pick-off signal noise: less than 0.01 milliradian peak-to-peak.

e. Characteristic time constant: less than 1.5 milliseconds.

f. Torque gain error: 0.5% for rates up to 500 mr per second

3.5.1.4.2 Gyro Excitation. The gyro spin motor and pick-off excitation supplied by the tracker power supply shall be a sine wave. In addition, the gyro pick-off excitation frequency shall be a multiple of gyro spin motor excitation frequency.

3.5.1.4.3 Gyro Alignment Accuracy. Each gimbal of the three-axis tracker shall be stabilized by single-axis gyroscopic inertial angular rate sensors. The azimuth and elevation sensors shall also be used to measure the azimuth and elevation LOS angular rates. The azimuth gyro input axis shall be aligned to within 5 mr of being orthogonal to the elevation gimbal axis. The elevation gyro input axis shall be aligned to within 10 mr of being parallel to the elevation gimbal axis when the azimuth gimbal angle is zero. The roll gyro input axis shall be aligned parallel to the roll gimbal axis within 20 mr.

3.5.1.5 Optical Sensor Alignment. The optical sensor horizontal scan lines shall be aligned to within one degree of the elevation axis of the

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gimbal platform. The LST detector shall be aligned in the optical axis so that the detector quadrants are "+", not "X". The LST detector LOS shall be aligned to the TV LOS to within ±0.5 mr.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all test requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests. Items covered by this specification shall be subjected to the tests listed in Table VI to determine compliance with all applicable requirements. (See Figure 6.)

a. First Article Tests (see 4.2).

b. Acceptance Tests (see 4.3).

c. System Performance Demonstration Tests (see 4.4).

d. Special Tests (see 4.5).

4.1.2 Special Working Environment. The contractor shall provide adequate facilities for the fabrication, assembly and testing of the items covered by this specification, in accordance with MIL-A-85055.

4.1.3 Government Verifications. Government verification shall be in accordance with MIL-A-85055.

4.1.4 Test Conditions. Unless otherwise specified in the contract, all tests for acceptance shall be performed at the standard conditions specfied by MIL-A-85055.

4.1.5 Spare Equipment Provisioning (WRA/SRA) Acceptance. Unless otherwise specified by the contract, it is intended that the items covered by this specification shall be accepted at the subsystem (DMT)/system (ARBS) level of assembly, in accordance with MIL-A-85055, to ensure that the equipment interchangeability and maintainability requirements of MIL-A-85055 are met.

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TABLE VI. Quality Assurance Test Requirements.

REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	TITLE	PREPRODUCTION TESTS	ACCEPTANCE TESTS
3.3.9	4.2.2.1, 4.6.1	Environmental Tests	X	
3.3.2 MIL-A-85055	4.2.2.2	Reliability Assurance Tests	X	
3.3.5	4.2.2.3	Electro-magnetic Interface Tests	X	
3.3.4, 3.3.6 MIL-A-85055	4.2.2.4	Maintainability Demo (I-Level)	X	
3.2, 3.3	4.3.1.1	Examination of Product	X	X
MIL-A-85055	4.3.1.2	Thermal Screening	X	X
MIL-A-85055	4.3.1.3	Random Vibration	X	X
3.4, 3.5	4.3.1.4, 4.6.2	Operational Tests	X	X
OPTIONAL	4.3.1.5	Optional Mfr's Burn-In	X	X
3.3.2	4.3.2	Reliability Assurance Tests		X
MIL-A-85055	4.4	System Tests	X	X
MIL-A-85055	4.5	Special Tests (AR=As Req'd)	AR	AR

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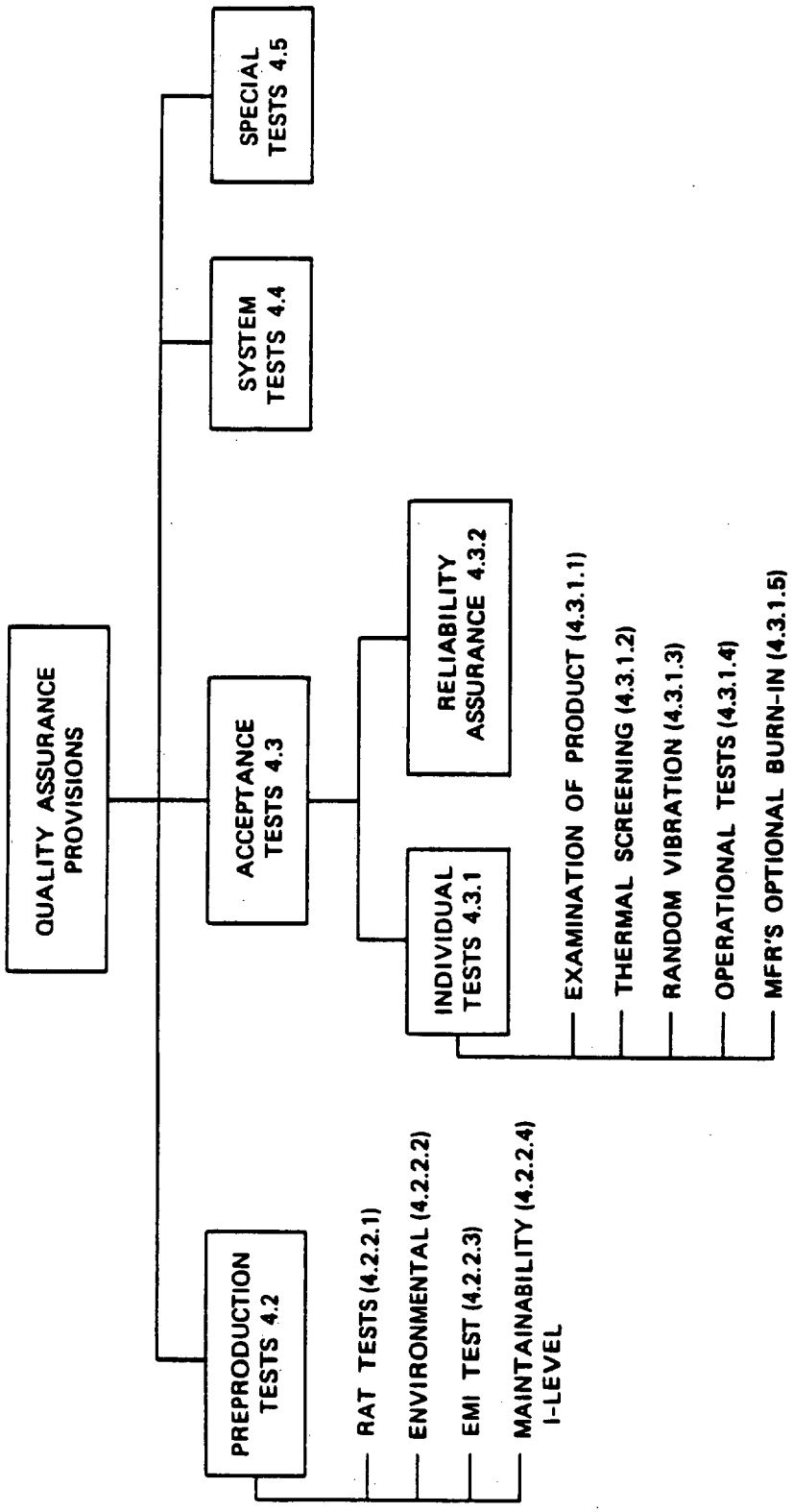


FIGURE 6. Classification of Tests.

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4.1.6 Manufacturing Reliability Screening. Each item covered by this specification shall be subjected to manufacturing reliability screening for the purpose of screening defective workmanship and parts, in accordance with MIL-A-85055.

4.2 First Article Tests. Preproduction tests shall be conducted by the contractor on equipment representative of the production equipment to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a Government representative may be designated to witness or supervise the test when so desired.
Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 First Article Test Data. The contractor shall make available all data collected in conducting these tests to the procuring activity for review in accordance with MIL-A-85055.

4.2.2 Scope of Tests. Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include:

a. Individual Tests (see 4.3.1). A-4M and AV-8B

b. Reliability Assurance Tests (see 4.2.2.1). A-4M only

c. Environmental Tests (see 4.2.2.2, 4.6.1). A-4M only

d. Electromagnetic Interference Tests (see 4.2.2.4). A-4M only

e. Maintainability Demonstration I-Level (see 4.2.2.4). A-4M only

f. Shock Test (see 4.6.1.8). AV-8B and A-4M

4.2.2.1 Preproduction Reliability Assurance Tests (RAT). Preproduction RATs shall be conducted in accordance with MIL-A-85055.

4.2.2.2 Environmental Tests. Environmental tests shall be performed in accordance with the procedure requirements of 4.6.1 to ensure that

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the requirements of 3.3.9, 3.3.10, 3.3.11 and 3.3.12 are met. The environmental test requirements are listed in Table VII. The test sequence shall be determined by the contractor, subject to approval by the Government inspector.

4.2.2.3 Electromagnetic Interference (EMI) Tests. EMI tests shall be performed in accordance with MIL-A-85055.

4.2.2.4 Maintainability Demonstration Test (I-Level). The first level (I-level) maintainability requirements specified by 3.3.4, 3.3.6 and MIL-A-85055 shall be verified by a maintainability test. The test requirements and procedures shall be prepared by the contractor and approved by the procuring activity. The test procedures shall inlcude procedures and requirements for adjustments and alignments required at all levels of assembly.

4.2.3 Preproduction (First Article) Approval. Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production trackers shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment to the approval of the preproduction sample is at the contractor's risk. Disposition of the preproduction sample shall be as specified in the contract or purchase order (see 6.2).

4.2.4 Production Trackers. Trackers supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each tracker shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection. For trackers already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Acceptance Tests. The contractor shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall prepare test records showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable, and shall be available for review upon request by the procuring activity. Acceptance or approval of material during the

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TABLE VII. Environmental Test Requirements.

REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	TITLE
3.3.9.1, 3.3.9.2	4.6.1.1	Low Temperature
3.3.9.1, 3.3.9.2	4.6.1.2	High Temperature
3.3.9.1	4.6.1.3	Temperature Shock
3.3.9.2	4.6.1.4	Low Pressure
3.3.9.3	4.6.1.5	Humidity
3.3.9.4	4.6.1.6	Vibration
3.3.9.4	4.6.1.7	Low Temperature Vibration
3.3.9.5	4.6.1.8	Shock
3.3.9.6	4.6.1.9	Gunfire Vibration
3.3.9.7	4.6.1.10	Acoustic Noise
3.3.9.8	4.6.1.11	Salt Spray
3.3.9.9	4.6.1.12	Sand and Dust
3.3.9.10	4.6.1.13	Explosive Atmosphere
3.3.9.11	4.6.1.14	Fungus
3.3.10	4.6.1.15	Warm-up Time
3.3.11	4.6.1 16	Operating Power

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course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests (see 4.3.1).

b. Reliability Assurance Tests (see 4.3.2).

4.3.1 Individual Tests. Each tracker submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each tracker accepted shall have passed the following tests:

a. Examination of Product (see 4.3.1.1).

b. Thermal Screening (see 4.3.1.2).

c. Random Vibration (see 4.3.1.3).

d. Operational tests (see 4.3.1.4).

e. Optional Manufacturer's Burn-in (see 4.3.1.5).

4.3.1.1 Examination of Product. Each tracker shall be examined carefully to determine that the material and workmanship requirements have been met. The tests shall consist of the physical and mechanical examinations listed in Table VIII.

4.3.1.2 Thermal Screening. Each item covered by this specification shall be subjected to thermal screening tests as specified by MIL-A-85055.

4.3.1.3 Random Vibration Tests. Each item covered by this specification shall be subjected to random vibration tests in accordance with MIL-A-85055 for the purpose of screening workmanship defects. The tests shall be performed in accordance with tests procedures prepared by the contractor and approved by the procuring activity.

4.3.1.4 Operational Tests. The tracker shall be subjected to the operational test specified in Table IX, in accordance with test procedures prepared by the contractor and approved by the procuring activity. The tests procedure requirements are outlined in 4.6.2. During the operational test, each unit under test shall be operated long enough to permit temperature stabilization, to provide sufficient time to check applicable characteristics, and to record adequate data to assure satis-

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TABLE VIII. Examination of Product Requirements.

REQUIREMENT PARAGRAPH	TITLE	PREPRODUCTION ACCEPTANCE	PRODUCTION ACCEPTANCE
MIL-A-85055			
3.2.4	Maintainability Provisions	X	
3.3	Design/Construction (MIL-E-5400)	X	
3.3.1	Parts and Materials (MIL-E-5400)	X	
3.3.1.2	Connectors	X	X
3.3.1.3	Adjustment Controls	X	
3.3.1.4	Printed Wiring Boards	X	X
3.3.1.5	Printed Wiring Assemblies	X	X
3.3.1.6	Elastomeric Materials	X	
3.3.1.7	Soldering	X	X
3.3.1.8	Wiring/Wire Coding	X	
3.3.1.9	Surface Finishes	X	X
3.3.1.10	Logic Circuits	X	
3.3.1.11	Screws and Fasteners	X	X
3.3.1.12	Vibration Isolation	X	
3.3.1.13	Accessibility	X	
3.3.2.5	Bonding and Grounding	X	X
3.3.3	Nomenclature/Nameplates/ID	X	X
3.3.3.1	Serialization	X	X
3.3.4	Workmanship	X	X
3.3.4.1	Cleanup	X	X
3.3.5	Interchangeability	X	

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TABLE VIII. Examination of Product Requirements (Continued).

REQUIREMENT PARAGRAPH	TITLE	PREPRODUCTION ACCEPTANCE	PRODUCTION ACCEPTANCE
MIL-A-85055 (CONT)			
3.3.8	Modules	X	
3.3.10	Cooling	X	
3.3.12	Boresight Provisions	X	
MIL-D-85056			
3.3.1.1	Total Weight	X	
3.3.1.2	Dimensions	X	X
3.3.2.4	Time Totalizing Meter	X	X
3.3.3.1	TM Capability	X	

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TABLE IX. Quality Assurance Operational (Performance) Tests.

REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	TITLE	PREPRODUCTION ACCEPTANCE	PRODUCTION ACCEPTANCE
3.4.6.1.3	4.6.2.1	Target Tracking	X	X
3.4.6.1.3.1	4.6.2.1	Tracking Rates	X	X
3.4.6.1.3.2	4.6.2.2	Tracking Rate Accuracy	X	X
3.4.6.1.3.3	4.6.2.3	Tracking Rate Noise	X	X
3.4.6.1.4	4.6.2.4	Slew Rate	X	X
3.4.7.1	4.6.2.5	LST Sensitivity	X	X
3.4.7.2	4.6.2.6	LST Field-of-View	X	
3.4.7.3	4.6.2.7	LST Off-axis Target Sensitivity	X	X
3.4.7.4	4.6.2.8	Laser Signal Processing	X	
3.4.7.4.1	4.6.2.8	Background Illumination	X	
3.4.7.4.2	4.6.2.8	False Acquisition Rate	X	X
3.4.7.4.3 3.4.7.4.4	4.6.2.9	Dynamic Range/Missed Commands	X	X
3.4.7.4.5 3.4.7.5.6	4.6.2.10	LST Correlation/Decorrelation	X	X
3.4.7.4.7	4.6.2.11	LST Multiple Target Discrimination	X	X
3.4.7.4.8	4.6.2.12	Pulse-to-pulse Response	X	X
3.4.7.4.9	4.6.2.13	LST Target Return Overload Response	X	X
3.4.7.4.10	4.6.2.14	LST Sun Blanking/Rejection	X	
3.4.7.4.11	4.6.2.5, 4.6.2.8, 4.6.2.9, 4.6.2.11, 4.6.2.12	Noise Variable Compensator (Proof of design, demonstrated by compliance with 3.4.7.1, 3.4.7.4.2, 3.4.7.4.3, 3.4.7.4.4, 3.4.7.4.7, and 3.4.7.4.8)	X	

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TABLE IX. Quality Assurance Operational (Performance) Tests. (Continued)

REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	TITLE	PREPRODUCTION ACCEPTANCE	PRODUCTION ACCEPTANCE
3.4.8.1	4.6.2.15	TV Light Level Range	X	X
3.4.8.2	(4.6.2.1)	TV Target Tracking	X	X
3.4.8.3	4.6.2.16	Sun Shutter Operation	X	X
3.4.8.4	4.6.2.17	TV Flare Light Sensitivity	X	
3.4.8.5 3.4.8.6	4.6.2.18	TV Field-of-View and Gate Size	X	
3.4.8.7	4.6.2.19	TV Signal Processing (Proof of design, demonstrate compliance with requirements of 3.4.8.7.1, 3.4.8.7.2, 3.4.8.7.3, 3.4.8.7.4, 3.4.8.7.5, 3.4.8.8, 3.4.8.9, 3.4.8.10)	X	
3.4.9	4.6.2.20	Boresight Accuracy (Proof of Design)	X	
3.4.10	4.6.2.21	Cooling System Leak Rate	X	X
3.5.1	4.6.2.22	Receiver-Processor (Proof of Design compliance with all applicable requirements)	X	
3.5.1.1.2	4.6.2.23	Tracking Gimbal Freedom	X	X
3.5.1.1.3	4.6.2.24	Base Motion Disturbance/Response	X	
3.4.5.3	4.6.2.25	Multiplexed Bus Test (AV-8B only)	X	X

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factory operation. Acceptance operational tests shall be performed prior to and after the completion of the reliability assurance (all equipment burn-in) tests.

4.3.1.5 Optional Manufacturer's Burn-in. Optional burn-in tests may be performed by the contractor in preparation for the reliability assurance tests.

4.3.2 Reliability Assurance Tests (RAT). Each tracker submitted for acceptance shall be subjected to the hours of burn-in, and the test conditions specified by MIL-A-85055, with the last 24 consecutive hours of test being failure-free for acceptance, in accordance with MIL-A-85055. The all equipment burn-in test for the AV-8B tracker shall be 48 hours. The accept/reject criteria of MIL-STD-781, Test Plan (level) XXIX shall be used for acceptance. During RAT on AV-8B trackers, the Functional Test A of MIL-A-85055 shall be replaced with a functional test of the multiplexed bus interface. The tests shall verify that data communication on the redundant buses shall comply with MDC A5232 and ICD 75J875900. The AV-8B Functional Test A shall include error handling tests as well as proper operating condition. Therefore, the Functional Tests for the AV-8B tracker shall consist of a new Functional Test A plus Functional Tests B and C of MIL-A-85055.

4.4 System Integration Tests A-4M only. Each tracker submitted for acceptance shall be functionally tested, in accordance with 4.6.3.1. The functional tests shall consist of demonstration of target acquisition, lock-on, TV/LST tracking and ARBS system BIT mode operation.

4.5 Special Tests. Special tests shall be conducted as required by the procuring activity, in accordance with MIL-A-85055.

4.6 Test Procedures. The procedures used for conducting tests specified herein shall be prepared by the contractor and approved by the procuring activity. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures shall be provided and shall be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional test, when deemed necessary.

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4.6.1 Environmental Tests. The tracker shall be subjected to environmental tests in accordance with the environmental test plan and the procedures of MIL-STD-810, except as modified herein.

4.6.1.1 Low Temperature Test. The tracker shall be tested at low temperature in accordance with MIL-STD-810, Test Method 501, Procedure I, and shall meet the requirements of 3.3.9.2. The storage temperature shall be -54°C and the soak time shall be at least 4 hours. The operating temperature shall be -40°C. Following completion of step 5 of the low temperature test, the tracker shall be stabilized at 20°C and 0°C for periods of at least 1 hour. Tracking accuracy tests (see 4.6.2.2) shall be performed at each temperature, if feasible. The procedure shall then be resumed at step 6.

4.6.1.2 High Temperature Test. The tracker shall be tested at high temperatures in accordance with MIL-STD-810, Test Method 501, Procedure 1, except as modified herein and shall meet the requirements of 3.3.9.2. Mounting and heat sinking of the tracker shall be in accordance with aircraft mounting provisions. The ambient operating temperature shall be +45°C after the nonoperating exposure period. Following the completion of step 5 of the high temperature test, the chamber temperature shall be raised to +60°C for 12 minutes. At the end of the 12-minute exposure, the temperature in the test chamber shall be raised to +70°C for 6 minutes. At the end of this period, the chamber temperature shall be reduced to +60°C for an additonal 12 minutes. The procedure shall be resumed at step 6. Following the completion of step 5, the tracker temperature shall be stabilized at +45°C. Tracking accuracy tests shall be performed, if feasible, at +45°C and +60°C. Tracker functional operation shall be verified at +70°C. At the completion of step 7, the chamber temperature shall be returned to +45°C and allowed to stabilize. With the tracker operating and while tracking a nominal target, the chamber temperature shall be raised to +60°C for 12 minutes. After this period, the chamber temperature shall be raised to +80°C for 6 minutes. After this period, the chamber temperature shall be lowered to +60°C for 12 minutes. The test shall then be resumed at step 6 of the procedure. The nominal target described in MIL-T-85115 may be used as the target for this test.

4.6.1.3 Temperature Shock Test. The tracker shall be tested with temperature shock in accordance with MIL-STD-810, Method 503, Procedure I, except that the maximum temperature shall be +85°C and shall meet the requirements of 3.3.9.1.

4.6.1.4 Low Pressure Test. The tracker shall be tested at low pressure in accordance with MIL-STD-810, Test Method 500, Procedure II. The tracker shall be operated and functionally tested during this test, and shall be capable of meeting the requirements of 3.4 and 3.5. The

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rate of pressure increase following the test shall be 15 pounds per square inch (psi) per minute.

4.6.1.5 Humidity Test. The tracker shall be tested in accordance with MIL-STD-810, Test Method 507, Procedure I, as modified by the conditions of Table I for a minimum of 240 hours. After removal from the test chamber and return to room ambient conditons, the tracker shall be subjected to the operational tests of 4.6.2.

4.6.1.6 Vibration. The tracker shall be tested with random vibration in each of three orthogonal axes and shall meet the requirements of 3.3.9.4. The test shall be in accordance with MIL-STD-810, Figure 514.1-4, Test Curve AE with the time being 30 minutes per axis.

4.6.1.7 Low Temperature-Vibration. The tracker shall be tested at a combined low temperature and vibration environment. The test shall be conducted at -40°C with random vibration of 0.02 g^2/Hz in one axis for 10 minutes after the tracker has stabilized at -40°C. The tracker shall be turned on prior to the start of vibration to allow sufficient time to warm up in accordance with 3.3.10. After completion of the vibration, the tracker shall meet the requirements of 3.4 and 3.5. Functional tests (see 4.6.3) shall be performed during the test.

4.6.1.8 Shock Test. The tracker shall be tested with shock in accordance with MIL-STD-810, Method 516, Procedure I and shall meet the requirements of 3.3.9.5. The shock pulse shape shall be in accordance with Figure 516.2-2 of MIL-STD-810.

4.6.1.9 Gunfire. The tracker shall be tested in accordance with MIL-STD-810, Method 519.1,Procedure I, to determine conformance with 3.3.9.6.

4.6.1.10 Acoustic Noise. The tracker shall be tested with acoustic noise in accordance with Test Method 515 of MIL-STD-810 for Category A equipment, unless certification analysis, approved by the procuring activity, demonstrates compliance with 3.3.9.7. While tracking the BIT source target during the test, the TV monitor video display and the false correlation functions shall be monitored for compliance with 3.3.9.7.

4.6.1.11 Salt Spray Test. The tracker shall be tested in accordance with MIL-STD-810, Method 509, Procedure I.

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4.6.1.12 Sand and Dust. The tracker shall be tested in accordance with MIL-STD-810, Test Method 510, Procedure I and shall meet the requirements of 3.3.9.9.

4.6.1.13 Explosive Atmosphere. The tracker shall be tested in accordance with MIL-STD-810, Method 511, Procedure I, unless certification analysis, approved by the procuring activity, demonstrates compliance with 3.3.9.10.

4.6.1.14 Fungus Test. Fungus testing shall be in accordance with MIL-STD-810, Method 508, Procedure I, unless certification analysis, approved by the procuring activity, demonstrates compliance with 3.3.9.11.

4.6.1.15 Warm-up Time. The tracker shall be tested for compliance with the warm-up requirements of 3.3.10. The following operational tests shall be performed after the specified warm-up time period:

a. TV target tracking (selected worst case targets, using the procedure of 4.6.2.1).

b. LST Sensitivity (Procedure 4.6.2.5).

The tracker shall be functionally tested in accordance with 4.6.3 after a warm-up time period of 5 minutes starting from a cold-soak low temperature condition of -26°C, to verify that the requirement of 3.3.10.1 is met.

4.6.1.16 Operating Power. The tracker shall be functionally tested in accordance with 4.6.3 under worst case combinations of the following input power conditions to verify compliance with 3.3.11.1, 3.3.11.2 and MIL-STD-704:

a. High input voltage.

b. Low input voltage.

c. High power frequency.

d. Low power frequency.

e. Line power transient conditions (separate tests).

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4.6.2 Operational Tests. Preproduction (First Article) and production trackers submitted for acceptance shall be subjected to the operational tests listed in Table IX, in accordance with the following paragraphs.

4.6.2.1 Target Tracking. Each tracker submitted for acceptance shall demonstrate the capability of tracking all targets specified by 3.4.3 as defined by MIL-T-85115. This shall be demonstrated with each target moving at a rate equal to, or greater than, 1 degree per second over an angular travel of the full electrical gimbal limits, and shall be demonstrated in both the azimuth and elevation gimbal axes. Tracking of LST coded targets defined by MIL-T-85115 shall be demonstrated as follows:

a. By correlating on targets with two randomly selected pulse repetition frequency (PRF) codes from Band I and with two randomly selected from Band II when the inserted code corresponds to the target PRF.

b. By not correlating on targets when the target PRF is different from the inserted code by plus or minus one code number.

A code selection procedure shall be set up so that each successive unit tests codes not previously tested by previous units.

4.6.2.2 Tracking Rate Accuracy. Tracking rate accuracy shall be measured in both the LST and TV. Accuracy shall be determined by evaluating the rate error signals corrected for offset and gain when corrected for earth rotation. The equipment shall meet the requirements of 3.4.6.1.3.2 over the range of tracking rates specified in 3.4.6.1.3.1. This test shall be performed at 20 times the sensitivity level specified in MIL-T-85115 for LST tracking and with the nominal TV target of MIL-T-85115 for TV tracking.

4.6.2.3 LST/TV Tracking Noise. LST and TV angular rate error output noise shall be measured with a true rms meter while the tracker is tracking a stationary target. This test shall be performed at 20 times the sensitivity level specified in MIL-T-85115 for LST tracking and with the nominal TV target of MIL-T-85115 for TV tracking. The measurements shall conform to 3.4.6.1.3.3.

4.6.2.4 Slew Rate. A simulated computer input shall be used to demonstrate the slew rate capability of the tracker, and to verify that the requirements of 3.4.6.1.4 are met.

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4.6.2.5 LST Sensitivity. LST sensitivity shall be measured using a laser target simulator capable of providing targets as defined in MIL-T-85115 (nominal pulse width, and rise and fall times) with pulse sequence of: 3 pulses with the correct interpulse period, then miss 10 pulses, repeat this sequence at least 30 times. Measurements shall be taken with the target source at 63 ±1 mr above and 63 ±1 mr right relative to the seeker optical axis. Repeat the test for the same value of target source set above and left, below and right, and below and left of the seeker optical axis. The minimum intensity of the target source on which it is possible to correlate on 50 percent of pulse sequences shall be in accordance with the requirements of MIL-T-85115. These tests shall be run with and without background.

4.6.2.6 LST Field of View. The LST field of view shall be measured on the preproduction tracker sample to verify that the requirements of 3.4.7.2 are met.

4.6.2.7 LST Off-Axis Target Sensitivity. LST off-axis target sensitivity shall be measured using a target source as specified in MIL-T-85115 with an intensity nominally 1,000 times the specified threshold level. With the target out of the LST field of view, allow the target to move toward the LST field of view and verify the requirements of 3.4.7.3 are met.

4.6.2.8 Laser Signal Processing. The preproduction tracker sample shall be tested to verify that the requirements of 3.4.7.4 are met. Each tracker submitted for acceptance shall be tested by applying the background specified in 3.4.7.4.1, monitoring the presence of correlated target pulses, and verifying that the requirements of 3.4.7.4.2 are met. This requirement can be satisified by measuring the raw alarm repetition rate for a 10-second period and showing by analysis that the false alarm rate is met.

4.6.2.9 LST Dynamic Range/Missed Commands. LST dynamic range/ missed commands shall be measured using a target source as specified in MIL-T-85115 with an energy level nominally 2, 4, 10, 100, 1,000, 10,000 and 100,000 times that specified. Measurements shall be made with the source initially set at 35 ±5 mr with respect to the LOS zero reference axis. The LST shall meet the requirements of 3.4.7.4.3 at a minimum level of 2 times the specified threshold. The number of missed commands occurring during any 1-minute interval shall be measured to verify that the tracker meets the requirements of 3.4.7.4.4 at a minimum level of 4 times the specified threshold.

4.6.2.10 LST Correlation/Decorrelation. LST correlation/decorrelation shall be measured by applying LST targets with characteristics as specified in MIL-T-85115.

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4.6.2.11 LST Multiple Target Discrimination (MTD) Logic. LST MTD logic shall be measure using dual target sources which are spatially separated by 17.5 ±5 mr. Perform the following tests to verify the requirements of 3.4.7.4.7:

a. With the first target at 40 times threshold, the second target at twice threshold, and the two targets separated in time by 100 ±10 ns, verify that the tracker is tracking only the first target. Repeat this test with the first target at 100,000 times threshold and the second target at 5,000 times threshold.

b. Hold the intensity of the first target at 20 times threshold and increase the intensity of the second target until the tracker is tracking only the second target. Verify that the intensity of the second target is twice threshold ±20 percent. Repeat this test with the amplitude of the first target at 100,000 times threshold. The amplitude of the second target shall be 10,000 times threshold ±20 percent.

c. Set the intensity of the first target at 40 times threshold and the intensity of the second target at twice threshold and separate the two targets in time by 10 to 20 ns. Verify that the tracker tracks exclusively either the first or second target without switching between the two targets. Repeat this test with the first target at 1,000 times threshold and the second target at 100 times threshold and with the first target at 10,000 times threshold and the second at 10,000 times threshold.

NOTE: The tests outlined above may be performed using targets having intensities different from those given, provided their relative intensities are as specified.

4.6.2.12 Pulse to-Pulse Response. The LST pulse-to-pulse response shall be verified by applying a pulse train at any selected PRF consisting of three pulses with a suitable intensity, followed by a pulse three times greater, followed by three additional pulses with the original intensity, followed by a pulse one-ninth less to verify the capability to meet the requirement of 3.4.7.4.8.

4.6.2.13 LST Target Return Overload Response. LST overload response shall be measured using a target source as specified in MIL-T-85115 positioned to 35 ±5 mr above and 35 ±5 mr to the right of the seeker axis. Increase the target intensity from below the specified threshold level to 500 times that level in a step overload and verify that the requirements of 3.4.7.4.9 are met.

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4.6.2.14 Sun Rejection. A sun rejection test shall be performed at the subassembly level to verify the requirements of 3.4.7.4.10.

4.6.2.15 TV Light Level Range. TV light level range shall be shown by illuminating the target background first to nominally 20 foot-lamberts and then to nominally 10,000 foot-lamberts during qualification testing. During acceptance testing, the light level range may be reduced to a level of 40 to 4,000 foot-lamberts. The tracker shall be locked onto and tracking the target at both illumination levels to verify the requirements of 3.4.8.1. The TV target used for this test shall be a nominal target specified by MIL-T-85115.

4.6.2.16 Sun Shutter Operation. Sun shutter operation shall be verified at the subassembly level to verify the requirements of 3.4.8.3 using He-Ne laser energy to simulate the sun. With no laser energy directed into either sun sensor, the sun shutter shall remain open. With laser energy directed into either sun sensor, the sun shutter shall close. As an acceptable alternative the sun shutter may be functionally tested at the subsystem level.

4.6.2.17 TV Flare Light Sensitivity. The preproduction tracker sample shall be tested to verify compliance with the requirements of 3.4.8.4 in accordance with a test procedure approved by the procuring activity.

4.6.2.18 TV Field of View and Gate Size. The preproduction sample tracker shall be tested to verify that the TV acquisition gate size, TV tracking gate size and TV field of view comply with the requirements of 3.4.8.5 and 3.4.8.6.

4.6.2.19 TV Signal Processing. The preproduction sample tracker shall be tested to verify that the requirements of 3.4.8.7, 3.4.8.8 and 3.4.8.9 are met.

4.6.2.20 Boresight Accuracy. The preproduction sample tracker shall be tested at the subsystem/system level to verify that the tracker accurately responds to position command inputs from the ARBS computer, within the accuracy requirement of 3.4.9 at the commanded boresight position.

4.6.2.21 Cooling System Leak Rate. The tracker shall be tested for leak rate not to exceed a pressure drop of 2 pounds per square inch over

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a 24-hours period, starting at a nominal pressure of one atmosphere above ambient. This test shall be performed with the Receiver-Processor and Heat Exchanger connected as a subsystem.

4.6.2.22 Receiver-Processor (Proof of Design). The preproduction tracker sample shall be tested to verify that all applicable requirements of 3.5.1 are met. The contractor shall prepare the acceptance requirements and test procedures, including all adjustments and alignments required at all levels of assembly that are necessary for final assembly and acceptance of the tracker.

4.6.2.23 Tracking Gimbal Freedom. Each tracker submitted for acceptance shall be tested to verify compliance with the tracking gimbal freedom specified in 3.5.1.1.2. The nominal TV target defined in MIL-T-85115 may be used for this test.

4.6.2.24 Base Motion Disturbance/Response. The preproduction tracker sample shall demonstrate the capability of tracking a TV target in the presence of base motion rates as specified in 3.5.1.1.3. The nominal TV target defined in MIL-T-85115 may be used for this test. The tracked target may be stationary, with the base motion rates of 3.5.1.1.3 applied to the mounting base. As a minimum, the azimuth and elevation axes shall be tested. The tracking rate noise shall not exceed that specified in 3.4.6.1.3.3 when the base is subjected to 50 percent of the rates and accelerations specified by 3.5.1.1.3 in the azimuth and elevation axes; and the tracker shall not lose lock-on when the maximum base motion rates and accelerations are applied.

4.6.2.25 Multiplexed Bus Test AV-8B only. The preproduction and production trackers shall be tested to verify that data communication on the redundant multiplexed buses shall comply with MDC A5232 and ICD 75J87900. These tests shall include error conditions as well as proper operating conditions.

4.6.3 Tracker Functional Tests.

4.6.3.1 A-4M Functional Tests. When integrated with the associated ARBS equipment, the tracker shall be capable of performing the Functional Tests A, B and C specified by MIL-A-85055. At the subsystem (tracker) level, the tracker shall be capable of acquiring and tracking TV and LST targets and correctly responding to input moding logic, serial digital data inputs and analog input functions specified by 3.4.5 and 3.4.6.1.1.

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4.6.3.2 AV-8B Functional Tests. Functional Test A of MIL-A-85055 shall be replaced with a functional test of the multiplexed data bus, as outlined in 4.3.2. The tracker shall be capable of performing this new Functional Test A plus Functional Tests B and C of MIL-A-85055. The tracker shall be capable of acquiring and tracking TV and LST targets and correctly respond to input moding logic, serial digital data inputs and analog input functions specified by 3.4.5 and 3.4.6.1.

5. PACKAGING

5.1 General. The tracker shall be prepared for delivery in accordance with MIL-A-85055.

6. NOTES

6.1 Intended Use. The dual mode tracker is intended for use as a major subsystem of the ARBS as part of the A-4M/AV-8B aircraft weapon delivery system.

6.2 Ordering Data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number and date of this specification.

(2) Spare Equipment Provisions (See 4.1.5).

(3) Selection of applicable levels of packaging and packing (See 5.1).

(4) WRA/SRA acceptance requirements (See 4.1.5.1).

(5) Disposition of preproduction sample (See 4.2.3).

6.2.1 Data Requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below will be delivered by the contractor in accordance with the contract or purchase order requirements.

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Paragraph	Data Requirements Title	Applicable DID no.
	Drawings, Engineering & Associated Lists	DI-E-7031
	Technical Directives	DI-E-21335A
	Report, Technical Directive Kit Shipment	UDI-A-21394
	List, Serial Number Configuration	UDI-E-21227A
	List, Integrated Configuration	UDI-E-21280
	Configuration Management Plan	DI-E-2035
3.2.1	Approved Source/Quality Verified Product List	UDI-E-21246A
	Quality Assurance Program Status Report	UDI-R-21373A
	Quality Assurance Program Plan	UDI-R-21374A
	Engineering Change Proposals & Requests for Deviation & Waiver	DI-E-2037
3.2.1	Non-Standard Part Approval Request	DI-E-7028
3.2.1	Data, Microelectronic Dev. & Assy. Req.	UDI-E-21337
3.3.2	Production Reliability Program Plan	UDI-R-7079
3.3.2	Report, Failure Summary	UDI-R-21139
	Failure Report	UDI-R-21141A
	Reliability Prediction Report	DI-R-7082
3.3.6	Report, Reliability and Maintainability Plan	UDI-R-21135
	Report, Reliability and Maintainability Test Results	UDI-R-21136
	System Safety Program Plan (SSPP)	DI-H-7047
	System Safety Engineering Report	DI-H-7050
	Data Accession List/Internal Data	UDI-A-26486
	Presentation Material	UDI-A-26347
3.3.7	Request, Nomenclature & Identification Assignment & Approval	UDI-E-21336B
3.3.7	Request, Nomenclature & Identification (Assign or Confirm)	UDI-E-21336B
3.3.7	Request, Nomenclature & Identification (Prefix letters)	UDI-E-21336B

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Paragraph	Data Requirements Title	Applicable DID no.
	Specifications , Equipment	UDI-E-21342
	Reliability & Maintainability Status Report	UDI-R-21137
4.3	Acceptance Test Report	DI-T-3721A
4.6	Test Procedures	UDI-T-21347

6.3 Acronyms and Abbreviations. The following acronyms and abbreviations are used in conjunction with the Dual Mode Tracker.

AC	-	Alternating current
ADL	-	Armament datum line
ARBS	-	Angle rate bombing set
BIT	-	Built-in-test
BITE	-	Built-in-test equipment
BST	-	Boresight
CIU	-	Converter-Interface unit
DMT	-	Dual Mode Tracker
EMI	-	Electromagnetic interference
FIT	-	Fault isolation test
g	-	Gravity unit(s)
Hz	-	Hertz
K	-	Thousand
KIAS	-	Knots indicated airspeed
LOS	-	Line of sight
LST	-	Laser Spot Tracker
MC	-	Mission computer
mr	-	Milliradians
MTBF	-	Mean-time-between-failure
MTD	-	Multiple target discrimination
NA	-	Not applicable
PRF	-	Pulse repetition frequency
psi	-	Pounds per square inch
RAT	-	Reliability acceptance test
RIG	-	Rate integrating gyro
RKTS	-	Rockets
rms	-	Root mean square
SRA(s)	-	Shop replaceable assembly(ies)
STBY	-	Standby
TV	-	Television
V	-	Volts
VAC	-	Volts alternating current
WDC	-	Weapon delivery computer
WRA(s)	-	Weapon replaceable assembly(ies)
<	-	Less than

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6.4 Precedence of Documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence should apply:

a. Contract. The contract should have precedence over any specification.

b. System Specification. The system specification (MIL-A-85055) should have precedence over this specification.

c. This Specification. This specification should have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications, where applicable, should be specifically approved in writing by the procuring activity.

d. Referenced Specifications. Any referenced specification should have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications should apply to the extent specified.

6.5 Performance Objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which should be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval should be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

6.6 Type Designation. The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Associated Equipment. The tracker operates with the equipment listed in 1.3.

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6.8 Changes From Previous Issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to extensiveness of the changes.

Preparing activity:
Navy - AS

(Project 1280-N019)

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APPENDIX
Message fromats for the SP-1C or AYK-14 Computer to Dual Mode Tracker for the AN/ASB 19(V) Angle Rate Bombing Set

This appendix details the message formats used between the SP-1C Weapon Delivery Computer (WDC) to Dual Mode Tracker (DMT), as used in the A-4M aircraft, and between the AYK-14 Mission Computer (MC) and the Dual Mode Tracker (DMT), as used in the AV-8B aircraft.

The Table 30 series lists the A-4M Message Formats. The Table 40 series lists the AV-8B Message Formats.

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30.0 A-4M MESSAGE FORMATS

30.1 Weapon Delivery Computer to Dual Mode Tracker Message Format

(a) Data Word 1 – Tracker Control Discretes
(b) Data Word 2 – LST Code Discretes
(c) Data Word 3 – Azimuth Command
(d) Data Word 4 – Elevation Command

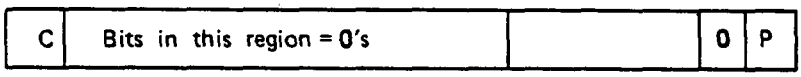
30.2 Dual Mode Tracker to Weapon Delivery Computer Message Format

(a) Data Word 1 – DMT Status Discretes
(b) Data Word 2 – Azimuth Gimbal Angle
(c) Data Word 3 – Elevation Gimbal Angle
(d) Data Word 4 – Roll Gimbal Angle
(e) Data Word 5 – Azimuth Rate Error
(f) Data Word 6 – Elevation Rate Error
(g) Data Word 7 – Elevation TV x-hair Error
(h) Data Word 8 – Azimuth TV x-hair Error
(i) Data Word 9 – Azimuth TV x-hair Rate
(j) Data Word 10 – Elevation TV x-hair Rate
(k) Data Word 11 – LST Azimuth Error
(l) Data Word 12 – LST Elevation Error

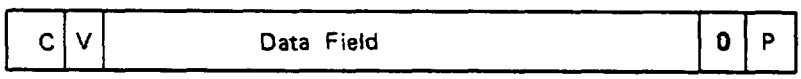
30.3 A-4M Address and Data Word Bit Patterns

C = Control Bit (=1) 0 = Logic Zero
V = Validity Bit (=1) 1 = Logic One
P = Parity Bit (Odd Ones)

Address
Word Bit No. 1 2 3 – – – – – – – – – – – – 15 16 17 18 19 20



Data
Word Bit No. 1 2 3 – – – – – – – – – – – – – – 18 19 20



Data
Field Bit No. 0 1 2 3 – – – – – – – – – – – – 14 15

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A-4M WDC to DMT Data Words

TABLE 30.1 (a)

Data Word 1 – Tracker Control Discretes

Bit No.	Title	Address Bit No.	15	16	17	18
0	Slew	Address Code	1	0	0	0
1	Lock-On					
2	Correlate					
3	Filter On					
4	Roll Reset					
5	Wide Gate					
6	FIT Pass					
7	Not Pull-Up Warning					
8	BIT Source On					
9	Yellow Filter Select					
10	Spare					
11	Spare					
12	Spare					
13	Linear Track Command					
14	Spare					
15	Spare					

TABLE 30.1 (b)

Data Word 2 – LST Code Discretes

Bit No.	Title	Address Bit No.	15	16	17	18
0	Spare	Address Code	0	1	0	0
1	Spare					
2	Spare					
3	Spare					
4	Spare					
5	Spare					
6	Spare					
7	LST Code 9					
8	LST Code 8					
9	LST Code 7					
10	LST Code 6					
11	LST Code 5					
12	LST Code 4					
13	LST Code 3					
14	LST Code 2					
15	LST Code 1					

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A-4M WDC to DMT Data Words

TABLE 30.1 (c)

Data Word 3 – Azimuth Command

Bit No.	Title	Address Bit No.	15	16	17	18
0	Spare	Address Code	1	1	0	0
1	Spare					
2	Spare					
3	Scale Factor (1 = High Rate, 0 = Low Rate)					

		Position Command Mode Scaling	Rate Command Mode Scaling (High Rate)	Rate Command Mode Scaling (Low Rate)
4	LSB	0.5 Sine of	250 mr/sec	62.5 mr/sec
5		0.25 Gimbal	125	31.25
6		0.125 Angle	62.5	15.625
7		0.0625	31.25	7.8125
8		0.03125	15.625	3.90625
9		0.015625	7.8125	1.95313
10		0.0078125	3.90625	0.97656
11		0.0039062	1.95313	0.48828
12		0.0019531	0.97656	0.24414
13		0.0009765	0.48828	0.12207
14	MSB	0.0004882	0.24414	0.06104
15		Sign (+ = Gimbal go Right)		

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A-4M WDC to DMT Data Words

TABLE 30.1 (d)

Data Word 4 — Elevation Command

Bit No.	Title	Address Bit No.	15	16	17	18
0	Spare	Address Code	0	0	0	0
1	Spare					
2	Spare					
3	Scale Factor (1 = High Rate, 0 = Low Rate)					

Bit No.		Position Command Mode Scaling	Rate Command Mode Scaling (High Rate)	Rate Command Mode Scaling (Low Rate)
4	LSB	0.5 Sine of	250 mr/sec	62.5 mr/sec
5		0.25 Gimbal	125	31.25
6		0.125 Angle	62.5	15.625
7		0.0625	31.25	7.8125
8		0.03125	15.625	3.90625
9		0.015625	7.8125	1.95313
10		0.0078125	3.90625	0.97656
11		0.0039062	1.95313	0.48828
12		0.0019531	0.97656	0.24414
13		0.0009765	0.48828	0.12207
14	MSB	0.0004882	0.24414	0.06104
15		Sign (+ = Gimbal go Up)		

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A-4M DMT to WDC Data Words

TABLE 30.2 (a)

Data Word 1 – DMT Status Discretes

Bit No.	Title	Address Bit No.	15	16	17	18
0	FIT A	Address Code	0	0	1	0
1	FIT B					
2	FIT C					
3	Linear Track					
4	Correlate Valid					
5	Pulse Valid					
6	Track Ready					
7	Rolled Over					
8	Over Heat					
9	Parity (Good)					
10	Not in FIT					
11	Not Under Heat					
12	Sun Shutter					
13	Good Signal					
14	Spare					
15	Spare					

Switch Position	FIT A	FIT B	FIT C
0	1	1	1
1	1	1	1
2	1	1	1
3	1	1	0
4	1	0	1
5	1	0	0
6	0	1	1
7	0	1	0
8	0	0	1
9	0	0	0

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A-4M DMT to WDC Data Words

TABLE 30.2 (b)

Data Word 2 – Azimuth Gimbal Angle

Bit No.	Title	Address Bit No.	15	16	17	18
0	Spare	Address Code	1	0	1	0
1	Spare					
2 LSB	0.02197265625 Degrees					
3	0.0439453125					
4	0.087890625					
5	0.17578125					
6	0.3515625					
7	0.703125					
8	1.40625					
9	2.8125					
10	5.625					
11	11.25					
12	22.5					
13	45.0					
14 MSB	90.0					
15	Sign (+ = Gimbal is Right)					

TABLE 30.2 (c)

Data Word 3 – Elevation Gimbal Angle

Bit No.	Title	Address Bit No.	15	16	17	18
0	Spare	Address Code	0	1	1	0
1	Spare					
2 LSB	0.02197265625 Degrees					
3	0.0439453125					
4	0.087890625					
5	0.17578125					
6	0.3515625					
7	0.703125					
8	1.40625					
9	2.8125					
10	5.625					
11	11.25					
12	22.5					
13	45.0					
14 MSB	90.0					
15	Sign (+ = Gimbal is Up)					

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A-4M DMT to WDC Data Words

TABLE 30.2 (d)

Data Word 4 – Roll Gimbal Angle

Bit No.	Title	Address Bit No.	15	16	17	18
0	Spare	Address Code	1	1	1	0
1	Spare					
2 LSB	0.02197265625 Degrees					
3	0.0439453125					
4	0.087890625					
5	0.17578125					
6	0.3515625					
7	0.703125					
8	1.40625					
9	2.8125					
10	5.625					
11	11.25					
12	22.5					
13	45.0					
14 MSB	90.0					
15	Sign (+ = CCW; Left Wing Down)					

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A-4M DMT to WDC Data Words

TABLE 30.2 (e)

Data Word 5 – Azimuth Rate Error

Bit No.	Title	Address Bit No.	15	16	17	18
0	Spare	Address Code	0	0	0	1
1	Spare					
2	Spare					
3	Spare					
4 LSB	0.0152587890625	Milliradians/Second				
5	0.030517578125					
6	0.06103515625					
7	0.1220703125					
8	0.244140625					
9	0.48828125					
10	0.9765625					
11	1.953125					
12	3.90625					
13	7.8125					
14 MSB	15.625					
15	Sign (+ = Target is Right)					

TABLE 30.2 (f)

Data Word 6 – Elevation Rate Error

Bit No.	Title	Address Bit No.	15	16	17	18
0	Spare	Address Code	1	0	0	1
1	Spare					
2	Spare					
3	Spare					
4 LSB	0.0152587890625	Milliradians/Second				
5	0.030517578125					
6	0.06103515625					
7	0.1220703125					
8	0.244140625					
9	0.48828125					
10	0.9765625					
11	1.953125					
12	3.90625					
13	7.8125					
14 MSB	15.625					
15	Sign (+ = Target is Up)					

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A-4M DMT to WDC Data Words

TABLE 30.2 (g)

Data Word 7 – Elevation TV x-hair Error

Bit No.		Title	Address Bit No.	15	16	17	18
0		Spare	Address Code	0	1	0	1
1		Spare					
2		Spare					
3		Spare					
4	LSB	0.0048828125 milliradians					
5		0.009765625					
6		0.01953125					
7		0.0390625					
8		0.078125					
9		0.15625					
10		0.3125					
11		0.625					
12		1.25					
13		2.5					
14	MSB	5.0					
15		Sign (+ = Target is Up)					

TABLE 30.2 (h)

Data Word 8 – Azimuth TV x-hair Error

Bit No.		Title	Address Bit No.	15	16	17	18
0		Spare	Address Code	1	1	0	1
1		Spare					
2		Spare					
3		Spare					
4	LSB	0.0048828125 milliradians					
5		0.009765625					
6		0.01953125					
7		0.0390625					
8		0.078125					
9		0.15625					
10		0.3125					
11		0.625					
12		1.25					
13		2.5					
14	MSB	5.0					
15		Sign (+ = Target is Right)					

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A-4M DMT to WDC Data Words

TABLE 30.2 (i)

Data Word 9 – Azimuth TV x-hair Rate				
Address Bit No.	15	16	17	18
Address Code	0	0	1	1
(GROWTH ITEM)				

TABLE 30.2 (j)

Data Word 10 – Elevation TV x-hair Rate				
Address Bit No.	15	16	17	18
Address Code	1	0	1	1
(GROWTH ITEM)				

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A-4M DMT to WDC Data Words

TABLE 30.2 (k)

Data Word 11 – LST Azimuth Error

Bit No.	Title	Address Bit No.	15	16	17	18
0	Spare	Address Code	0	1	1	1
1	Spare					
2	Spare					
3	Spare					
4 LSB	0.0016272996875 milliradians					
5	0.003255859375					
6	0.00651171875					
7	0.0130234375					
8	0.026046875					
9	0.05209375					
10	0.1041875					
11	0.208375					
12	0.41675					
13	0.8335					
14 MSB	1.667					
15	Sign (+ = Target is Right)					

TABLE 30.2 (l)

Data Word 12 – LST Elevation Error

Bit No.	Title	Address Bit No.	15	16	17	18
0	Spare	Address Code	1	1	1	1
1	Spare					
2	Spare					
3	Spare					
4 LSB	0.0016272996875 milliradians					
5	0.003255859375					
6	0.00651171875					
7	0.0130234375					
8	0.026046875					
9	0.05209375					
10	0.1041875					
11	0.208375					
12	0.41675					
13	0.8335					
14 MSB	1.667					
15	Sign (+ = Target is Up)					

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TABLE 40.0. AV-8B Message Formats.

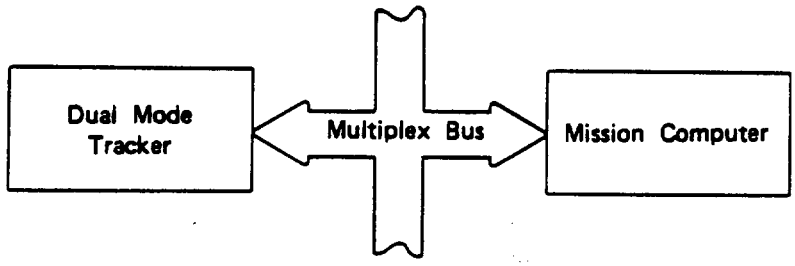


TABLE 40.1

40.1 Mission Computer (MC) to Dual Mode Tracker (DMT) Message Format

Message No. 1

a. Command Word from MC to DMT
b. Data Word 1 – Discrete Word 1
c. Data Word 2 – Discrete Word 2
d. Data Word 3 – AZ Command
e. Data Word 4 – EL Command
f. Status Word from DMT to MC

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TABLE 40.1 (a). AV-8B Message Formats.

Message No. 1 (Data from MC to DMT)

Terminal Address	T/R	Subaddress/ Mode	Word Count
0 1 1 0 0	0	0 0 0 0 1	0 0 1 0 0

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Command Word to DMT

Word No.	
1	Discrete Word 1 – Tracker Control
2	Discrete Word 2 – LST Code
3	Azimuth Command
4	Elevation Command

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TABLE 40.1 (b). AV-8B Message Formats.

Message No. 1 (MC to DMT)

Tracker Control	Not Used

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 1

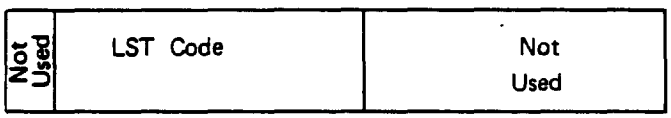
0 Slew
1 TV Lock-On
2 LST Correlate
3 Analog Filter On
4 Roll Reset
5 Wide Gate
6 FIT Pass
7 Not Pull-Up Warning
8 BIT Source On
9 Yellow Filter Select (Red = 0, Yel = 1)
10 Linear Track Command
11 Mode A
12 Mode B } DMT Mode Control – See Table Below
13 Mode C
14 Spare
15 Spare

Mode A	Mode B	Mode C	DMT Mode
1	1	1	Off
1	1	0	WDC
1	0	1	STBY
1	0	0	TV
0	1	1	Spare
0	1	0	LST
0	0	1	BST
0	0	0	BIT

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TABLE 40.1 (c). AV-8B Message Formats.

Message No. 1 (MC to DMT)



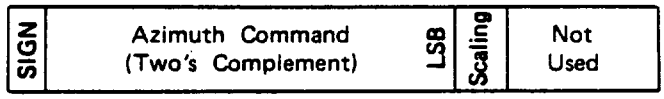
Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 2	
0	Spare
1	LST Code 9
2	LST Code 8
3	LST Code 7
4	LST Code 6
5	LST Code 5
6	LST Code 4
7	LST Code 3
8	LST Code 2
9	LST Code 1
10	Spare
11	Spare
12	Spare
13	Spare
14	Spare
15	Spare

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TABLE 40.1 (d). AV-8B Message Formats.

Message No. 1 (MC to DMT)



Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 3

0 Sign (+ = Gimbal go Right)

Position Command Mode

Rate Command Mode

			High Rate (Scaling BIT = 1)	Low Rate (Scaling BIT = 0)
1	LSB	0.5 Sine of	250 mr/sec	62.5 mr/sec
2		0.25 Gimbal	125	31.25
3		0.125 Angle	62.5	15.625
4		0.0625	31.25	7.8125
5		0.03125	15.625	3.90625
6		0.015625	7.8125	1.95313
7		0.0078125	3.90625	0.97656
8		0.0039062	1.95313	0.48828
9		0.0019531	0.97656	0.24414
10		0.0009765	0.48828	0.12207
11	MSB	0.0004882	0.24414	0.06104

12 Scaling (1 = High Rate, 0 = Low Rate)

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TABLE 40.1 (e). AV-8B Message Formats.

Message No. 1 (MC to DMT)

SIGN	MSB	Elevation Command (Two's Complement)	LSB	Scaling	Not Used

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 4

0 Sign (+ = Gimbal go Up)

Position Command Mode

Rate Command Mode

			High Rate (Scaling BIT = 1)	Low Rate (Scaling BIT = 0)
1	LSB	0.5 Sine of	250 mr/sec	62.5 mr/sec
2		0.25 Gimbal	125	31.25
3		0.125 Angle	62.5	15.625
4		0.0625	31.25	7.8125
5		0.03125	15.625	3.90625
6		0.015625	7.8125	1.95313
7		0.0078125	3.90625	0.97656
8		0.0039062	1.95313	0.48828
9		0.0019531	0.97656	0.24414
10		0.0009765	0.48828	0.12207
11	MSB	0.0004882	0.24414	0.06104

12 Scaling (1 = High Rate, 0 = Low Rate)

TABLE 40.1 (f). AV-8B Message Formats.

Message No. 1

Terminal Address	ME	Status	TF
0 1 1 0 0		0 0 0 0 0 0 0 0 0	0

ME = Message Error
TF = Terminal Flag

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Status Word From DMT to MC

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TABLE 40.2

40.2 Dual Mode Tracker (DMT) to Mission Computer (MC) Message Format

Message No. 2

a. Command Word from MC to DMT
b. Status Word from DMT to MC
c. Data Word 1 – Discrete Word 1
d. Data Word 2 – Azimuth Gimbal Angle
e. Data Word 3 – Elevation Gimbal Angle
f. Data Word 4 – Roll Gimbal Angle
g. Data Word 5 – Azimuth Rate Error
h. Data Word 6 – Elevation Rate Error
i. Data Word 7 – Elevation TV x-hair Error
j. Data Word 8 – Azimuth TV x-hair Error
k. Data Word 9 – Azimuth TV x-hair Rate
l. Data Word 10 – Elevation TV x-hair Rate
m. Data Word 11 – LST Azimuth Error
n. Data Word 12 – LST Elevation Error

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Check the source to verify that this is the current version before use.

TABLE 40.2 (a).

Terminal Address	T/R	Subaddress/ Mode	Word Count
0 1 1 0 0	1	0 0 0 1 0	0 1 1 0 0

Bit No.	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Command Word From MC to DMT

Word No.	Parameter
1	Discrete Word 1
2	Azimuth Gimbal Angle
3	Elevation Gimbal Angle
4	Roll Gimbal Angle
5	Azimuth Rate Error
6	Elevation Rate Error
7	Elevation TV x-hair Error
8	Azimuth TV x-hair Error
9	Azimuth TV x-hair Rate
10	Elevation TV x-hair Rate
11	LST Azimuth Error
12	LST Elevation Error

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TABLE 40.2 (b). AV-8B Message Formats.

Message No. 2 (Data From DMT to MC)

Terminal Address	ME	Status	TF
0 1 1 0 0		0 0 0 0 0 0 0 0 0	0

ME = Message Error
TF = Terminal Flag

Bit No.	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Status Word From DMT to MC

TABLE 40.2 (c). AV-8B Message Formats.

Message No. 2 (DMT to MC)

Tracker Reporting Discreste	Not Used

Bit No.	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Word No. 1

0	FIT A
1	FIT B
2	FIT C
3	Linear Track
4	Correlation Valid
5	Pulse Valid
6	Tracker Ready
7	Rolled Over
8	Overheat
9	Spare
10	Not in FIT
11	Not Underheat
12	Sun Shutter Monitor
13	Good Signal
14	Spare
15	Spare

Switch Pos.	FIT A	FIT B	FIT C
0	1	1	1
1	1	1	1
2	1	1	1
3	1	1	0
4	1	0	1
5	1	0	0
6	0	1	1
7	0	1	0
8	0	0	1
9	0	0	0

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TABLE 40.2 (d). AV-8B Message Formats.

Message No. 2 (DMT to MC)



Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 2		
0	Sign	(+ = Gimbal is Right)
1	MSB	90 Degrees
2		45
3		22.5
4		11.25
5		5.625
6		2.8125
7		1.40625
8		0.703125
9		0.3515625
10		0.17578125
11		0.087890625
12		0.0439453125
13	LSB	0.02197265625

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

TABLE 40.2 (e) AV-8B Message Formats.

Message No. 2 (DMT to MC)

SIGN	MSB	Elevation Gimbal Angle	LSB	Not Used

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 3

0	Sign	(+ = Gimbal is Up)
1	MSB	90 Degrees
2		45
3		22.5
4		11.25
5		5.625
6		2.8125
7		1.40625
8		0.703125
9		0.3515625
10		0.17578125
11		0.087890625
12		0.0439453125
13	LSB	0.02197265625

TABLE 40.2 (f). AV-8B Message Formats.

Message No. 2 (DMT to MC)

SIGN	MSB	Roll Gimbal Angle	LSB	Not Used

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 4

0	Sign	(+ = CCW; Left Wing Down)
1	MSB	90 Degrees
2		45
3		22.5
4		11.25
5		5.625
6		2.8125
7		1.40625
8		0.703125
9		0.3515625
10		0.17578125
11		0.987890625
12		0.0439453125
13	LSB	0.02197265625

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TABLE 40.2 (g). AV-8B Message Formats.

Message No. 2 (DMT to MC)

SIGN	Azimuth Rate Error	LSB	Not Used

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 5

0	Sign	(+ = Target is to Right)
1	MSB	15.625 mr/sec
2		7.8125
3		3.90625
4		1.953125
5		0.9765625
6		0.48828125
7		0.244140625
8		0.1220703125
9		0.06103515625
10		0.030517578125
11	LSB	0.0152587890625

TABLE 40.2 (h). AV-8B Message Formats.

Message No. 2 (DMT to MC)

SIGN	MSB	Elevation Rate Error	LSB	Not Used

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 6

0	Sign	(+ = Target is Up)
1	MSB	15.625 mr/sec
2		7.8125
3		3.90625
4		1.953125
5		0.9765625
6		0.48828125
7		0.244140625
8		0.1220703125
9		0.06103515625
10		0.030517579125
11	LSB	0.0152587890625

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TABLE 40.2 (i). AV-8B Message Formats.

Message No. 2 (DMT to MC)

SIGN	MSB	Elevation TV x-hair Error	LSB	Not Used

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 7

0	Sign	(+ = Target is Up)
1	MSB	5.0 mr
2		2.5
3		1.25
4		0.625
5		0.3125
6		0.15625
7		0.078125
8		0.0390625
9		0.01953125
10		0.009765625
11	LSB	0.0048828125

TABLE 40.2 (j). AV-8B Message Formats.

Message No. 2 (DMT to MC)

SIGN	MSB	Azimuth TV x-hair Error	LSB	Not Used

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 8

0	Sign	(+ = Target is Right)
1	MSB	5.0 mr
2		2.5
3		1.25
4		0.625
5		0.3125
6		0.15625
7		0.078125
8		0.0390625
9		0.01953125
10		0.009765625
11	LSB	0.0048828125

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TABLE 40.2 (k). AV-8B Message Formats.

Message No. 2 (DMT to MC)

SIGN	MSB	Azimuth TV x-hair Rate	LSB	Not Used

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 9
(Growth Item)

TABLE 40.2 (l). AV-8B Message Formats.

Message No. 2 (DMT to MC)

SIGN	MSB	Elevation TV x-hair Rate	LSB	Not Used

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 10
(Growth Item)

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TABLE 40.2 (m). AV-8B Message Formats.

Message No. 2 (DMT to MC)

SIGN	MSB	LST Azimuth Error	LSB	Not Used

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 11

0	Sign	(+ = Target is Right)
1	MSB	1.667 mr
2		0.8335
3		0.41675
4		0.208375
5		0.1041875
6		0.05209375
7		0.026046875
8		0.0130234375
9		0.00651171875
10		0.003255859375
11	LSB	0.0016279296875

TABLE 40.2 (n). AV-8B Message Formats.

Message No. 2 (DMT to MC)

SIGN	MSB	LST Elevation Error	LSB	Not Used

Bit No. 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Word No. 11

0	Sign	(+ = Target is Up)
1	MSB	1.667 mr
2		0.8335
3		0.41675
4		0.208375
5		0.1041875
6		0.05209375
7		0.026048875
8		0.0130234375
9		0.00651171875
10		0.003255659375
11		0.0016272996875

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